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(54) **CROSS-VALIDATING SENSORS OF AN AUTONOMOUS VEHICLE**

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(56) **References Cited**

U.S. PATENT DOCUMENTS

5,214,408 A * 5/1993 Asayama G01B 11/026
340/435
8,521,352 B1 * 8/2013 Ferguson G05D 1/0272
701/25

(Continued)

FOREIGN PATENT DOCUMENTS

JP H06195121 A 7/1994

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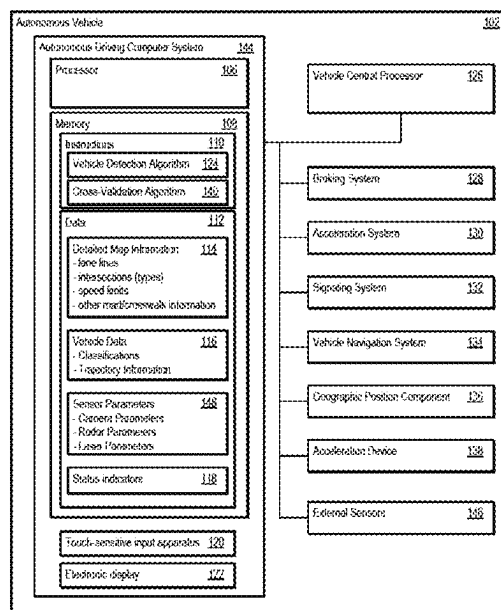
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(57) **ABSTRACT**

Methods and systems are disclosed for cross-validating a second sensor with a first sensor. Cross-validating the second sensor may include obtaining sensor readings from the first sensor and comparing the sensor readings from the first sensor with sensor readings obtained from the second sensor. In particular, the comparison of the sensor readings may include comparing state information about a vehicle detected by the first sensor and the second sensor. In addition, comparing the sensor readings may include obtaining a first image from the first sensor, obtaining a second image from the second sensor, and then comparing various characteristics of the images. One characteristic that may be compared are object labels applied to the vehicle detected by the first and second sensor. The first and second sensors may be different types of sensors.

20 Claims, 14 Drawing Sheets



Related U.S. Application Data

continuation of application No. 15/825,674, filed on Nov. 29, 2017, now Pat. No. 11,518,395, which is a continuation of application No. 15/383,198, filed on Dec. 19, 2016, now Pat. No. 9,868,446, which is a continuation of application No. 14/946,967, filed on Nov. 20, 2015, now Pat. No. 9,555,740, which is a continuation of application No. 13/628,986, filed on Sep. 27, 2012, now Pat. No. 9,221,396.

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(56) **References Cited**

U.S. PATENT DOCUMENTS

9,221,396 B1 12/2015 Zhu et al.
 9,868,446 B1 1/2018 Zhu et al.

2003/0209893 A1* 11/2003 Breed B60R 21/01554
 701/45
 2004/0258279 A1* 12/2004 Hirvonen G06V 20/64
 382/104
 2005/0131646 A1* 6/2005 Camus G06V 20/58
 382/104
 2006/0208169 A1* 9/2006 Breed G01S 17/88
 250/221
 2008/0012938 A1* 1/2008 Kubota B60Q 9/008
 348/118
 2009/0115610 A1* 5/2009 Steinhage G01S 1/68
 340/572.1
 2009/0177347 A1* 7/2009 Breuer B60W 50/0205
 701/31.4
 2010/0104199 A1* 4/2010 Zhang G08G 1/166
 382/199
 2010/0169013 A1* 7/2010 Nakamura G08G 1/0969
 701/469
 2010/0312461 A1* 12/2010 Haynie B61L 25/026
 701/117
 2011/0035123 A1* 2/2011 Katrak G01B 7/30
 701/55
 2011/0075518 A1* 3/2011 Huhta G01S 15/60
 367/90
 2011/0115615 A1* 5/2011 Luo B60R 1/30
 348/148
 2011/0118967 A1 5/2011 Tsuda
 2011/0166746 A1* 7/2011 Breuer B60T 17/221
 701/29.2
 2012/0136510 A1* 5/2012 Min G01S 17/88
 701/2
 2012/0269387 A1* 10/2012 Becker G06T 7/254
 382/103
 2013/0058527 A1* 3/2013 Peynot G01S 17/86
 382/103
 2013/0245877 A1* 9/2013 Ferguson G06V 20/56
 701/23
 2013/0300872 A1* 11/2013 Park G08G 1/167
 348/148
 2013/0321629 A1* 12/2013 Zhang G06T 3/00
 348/148
 2022/0358315 A1* 11/2022 Qiu G06V 20/56

* cited by examiner

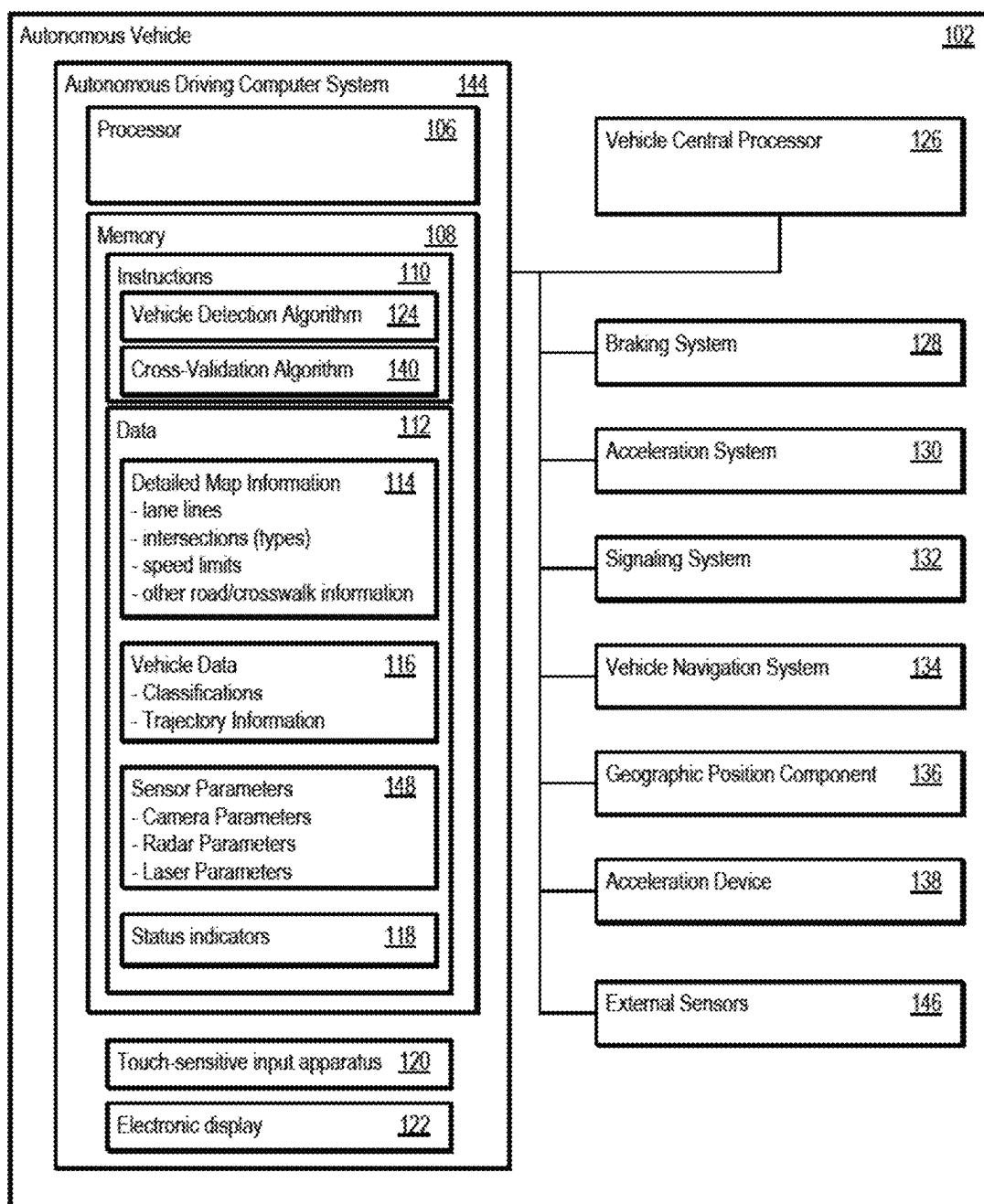


FIG. 1

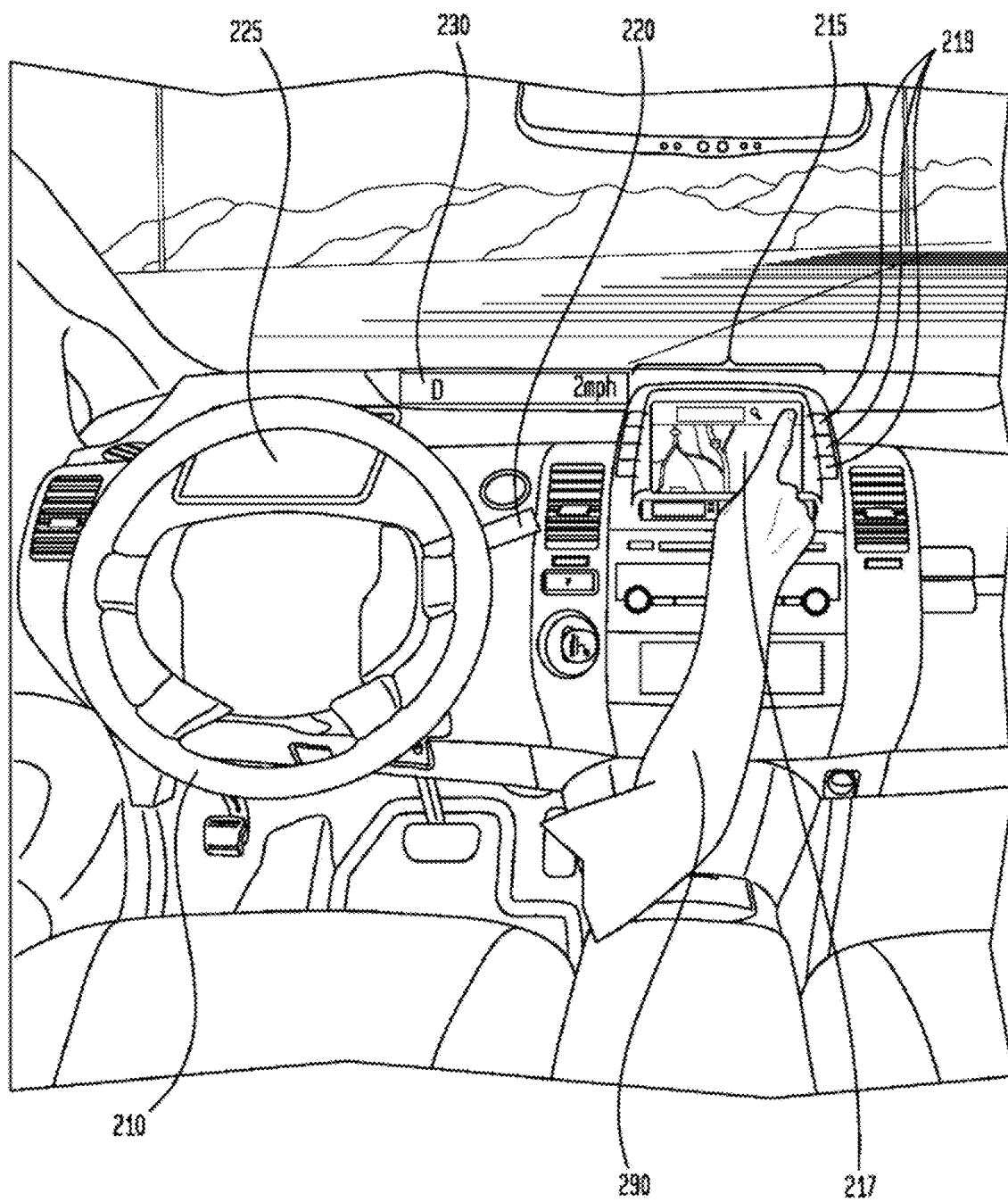


FIG. 2

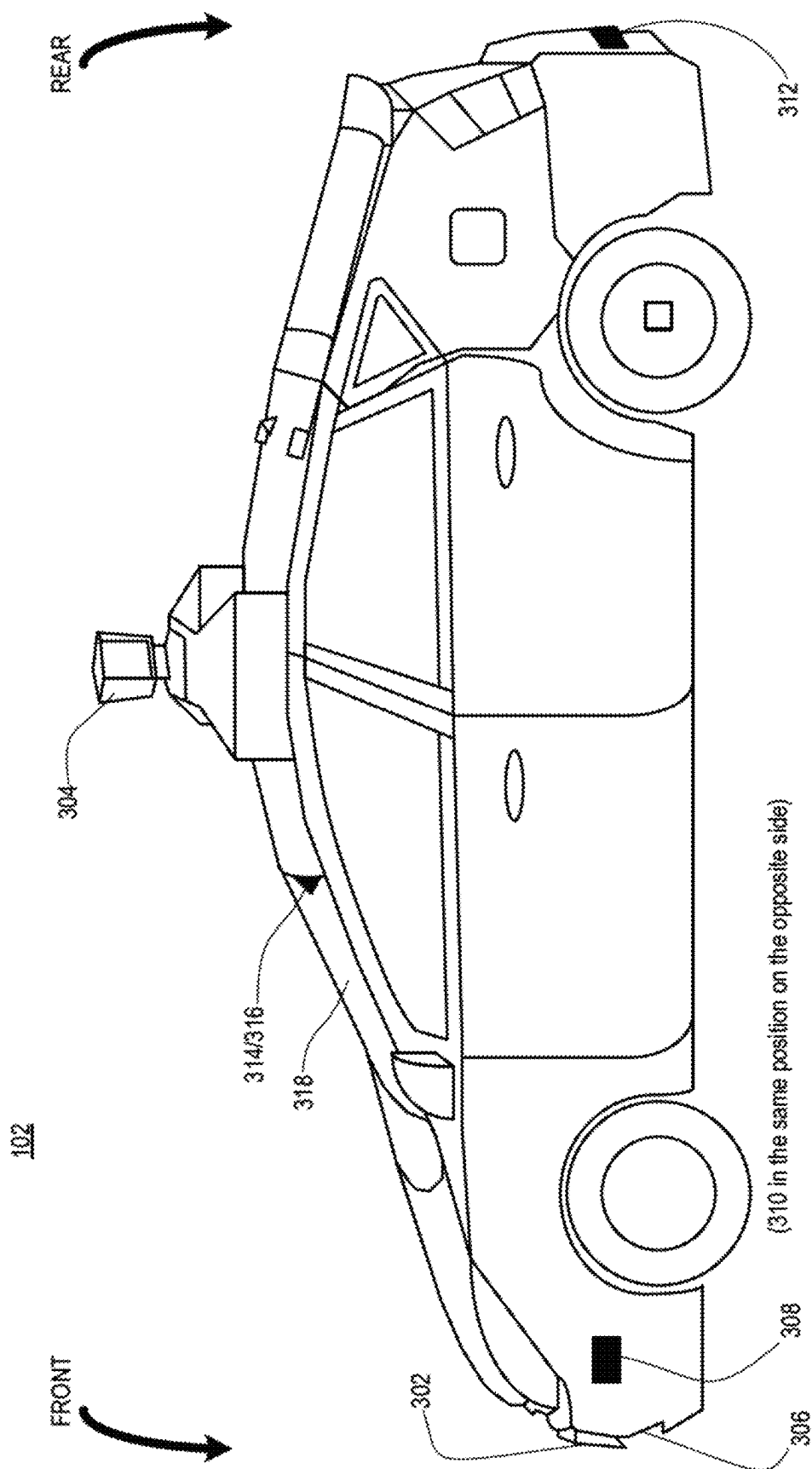


FIG. 3

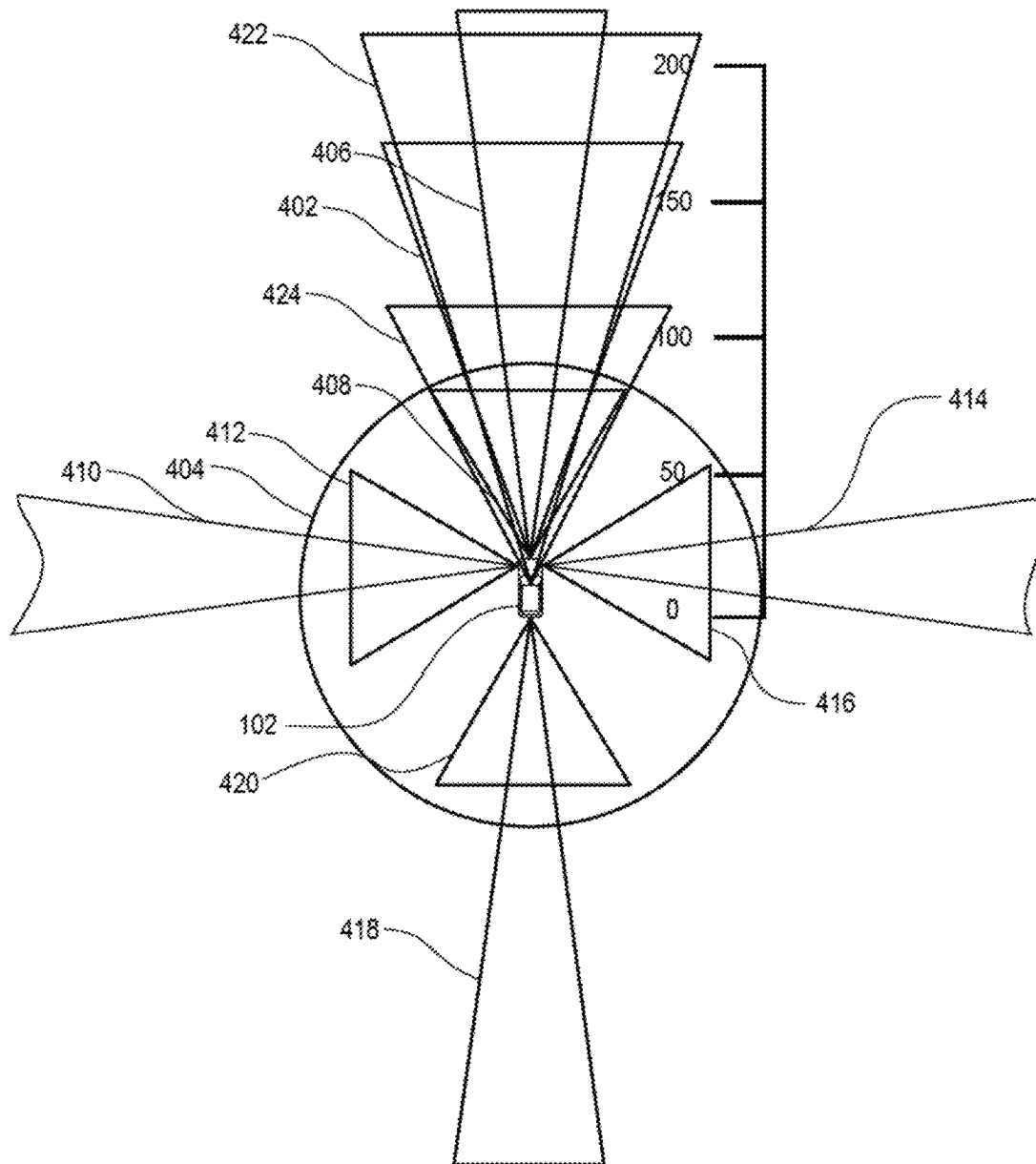


FIG. 4A

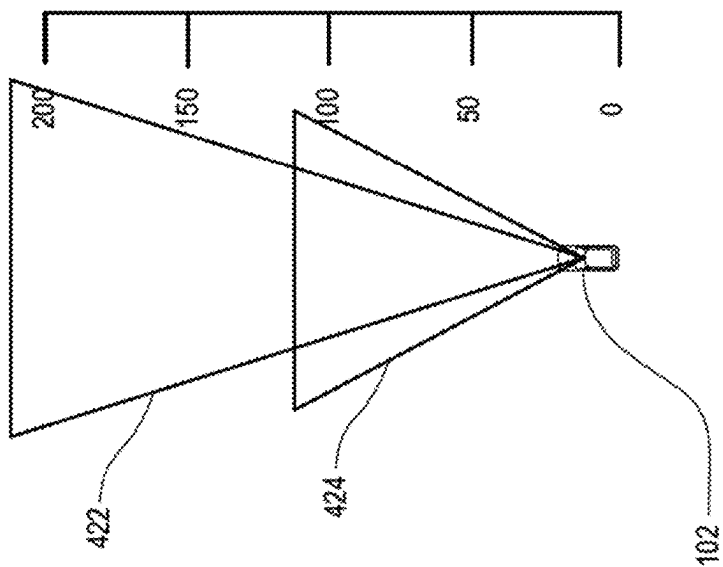


FIG. 4D

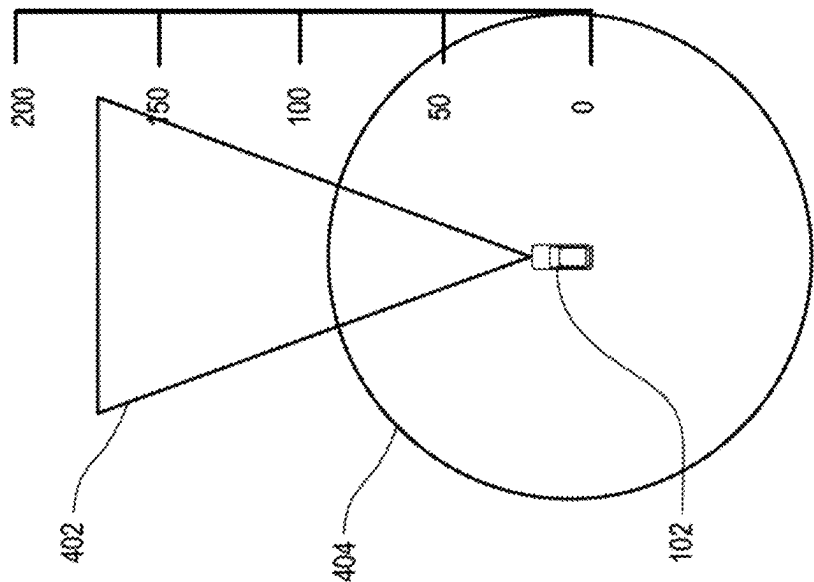


FIG. 4B

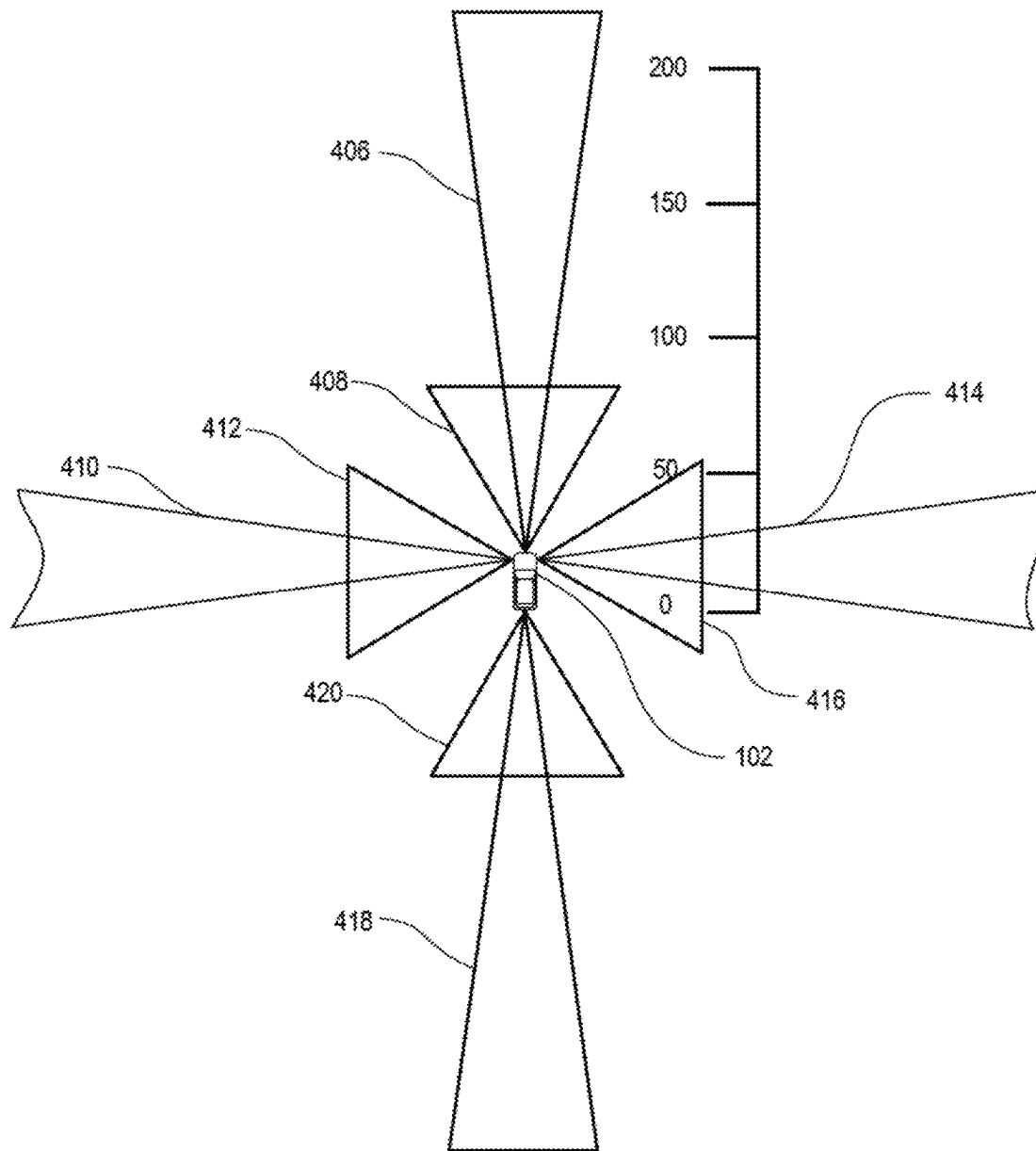


FIG. 4C

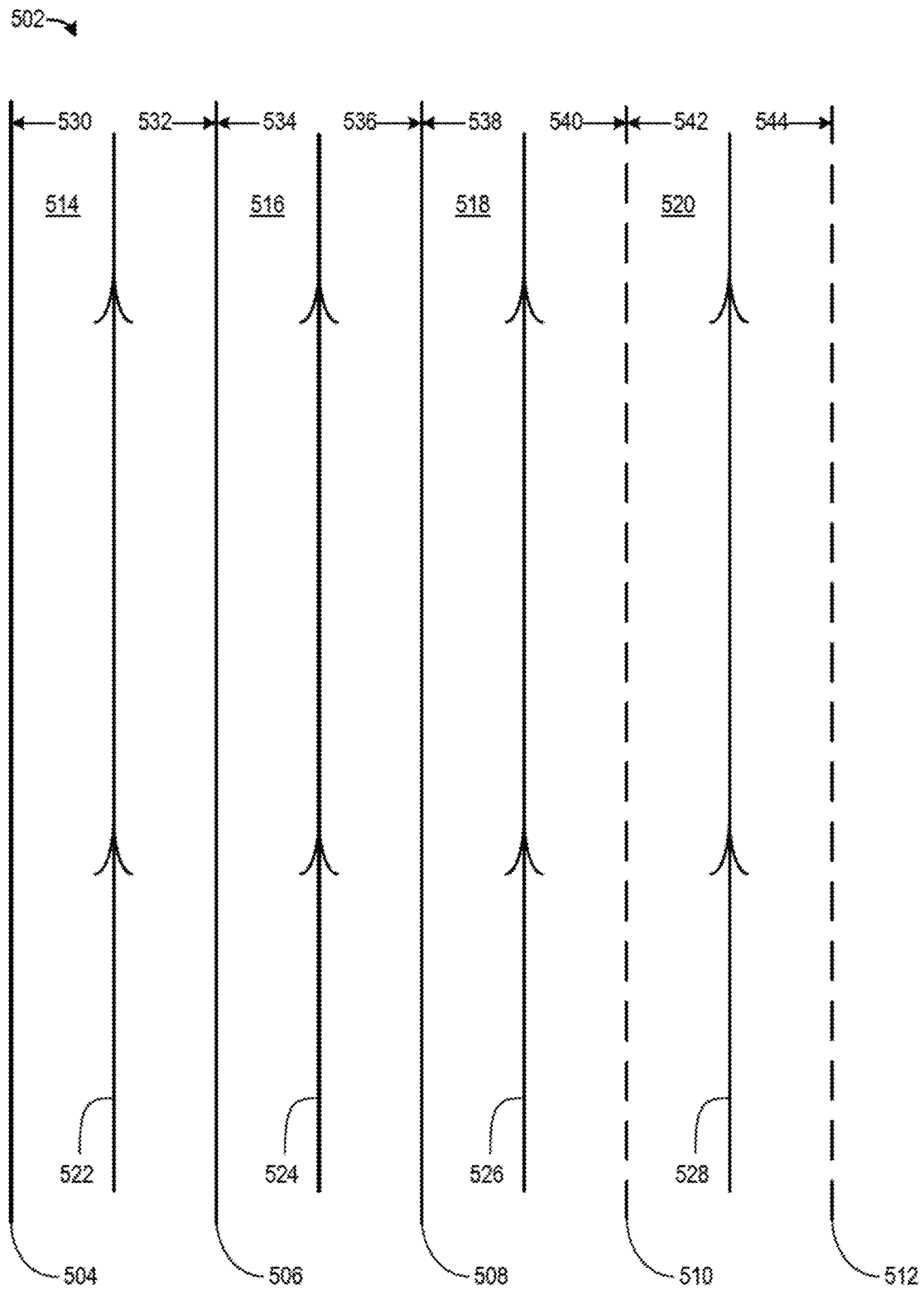


FIG. 5

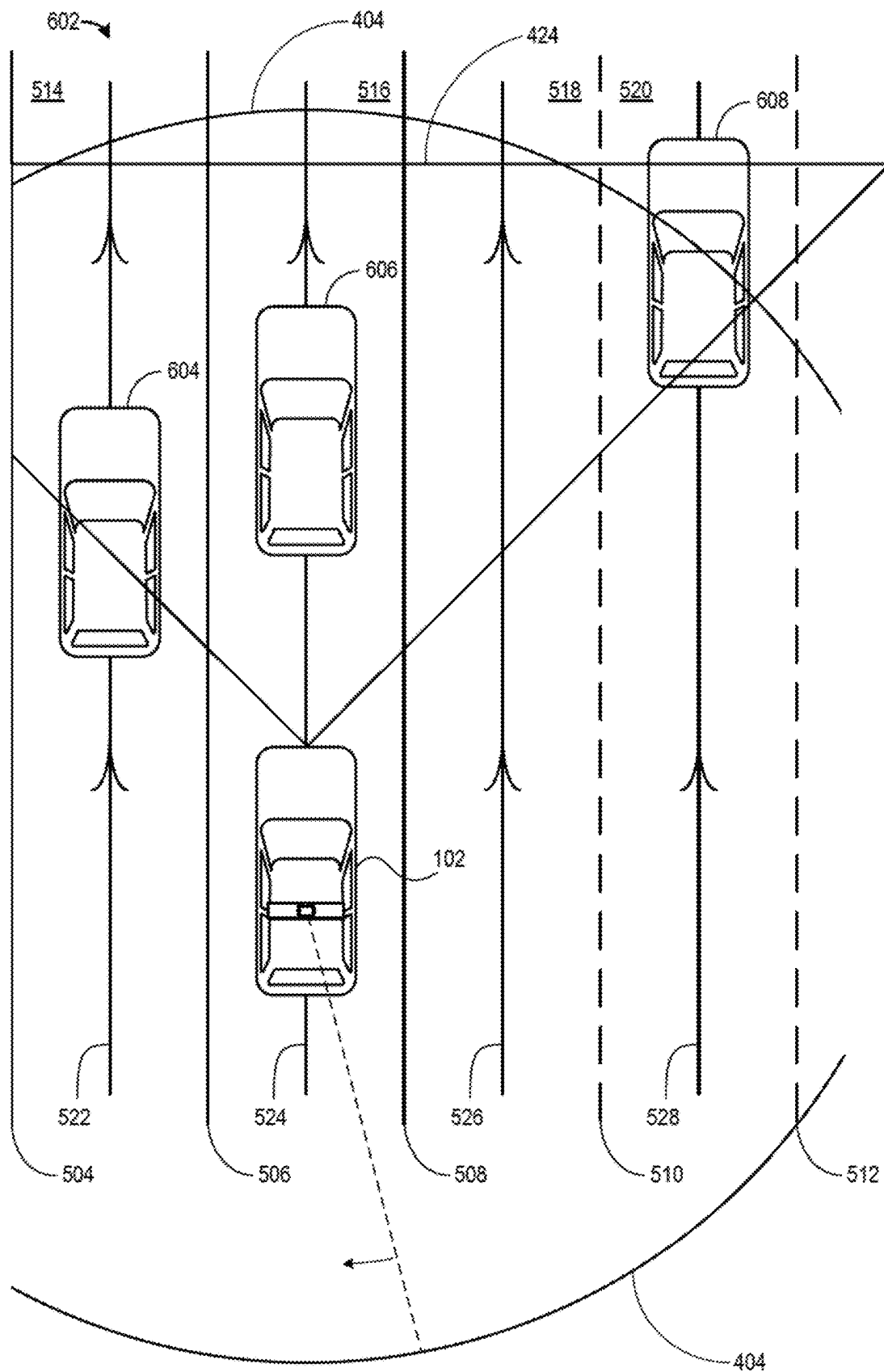


FIG. 6



FIG. 7

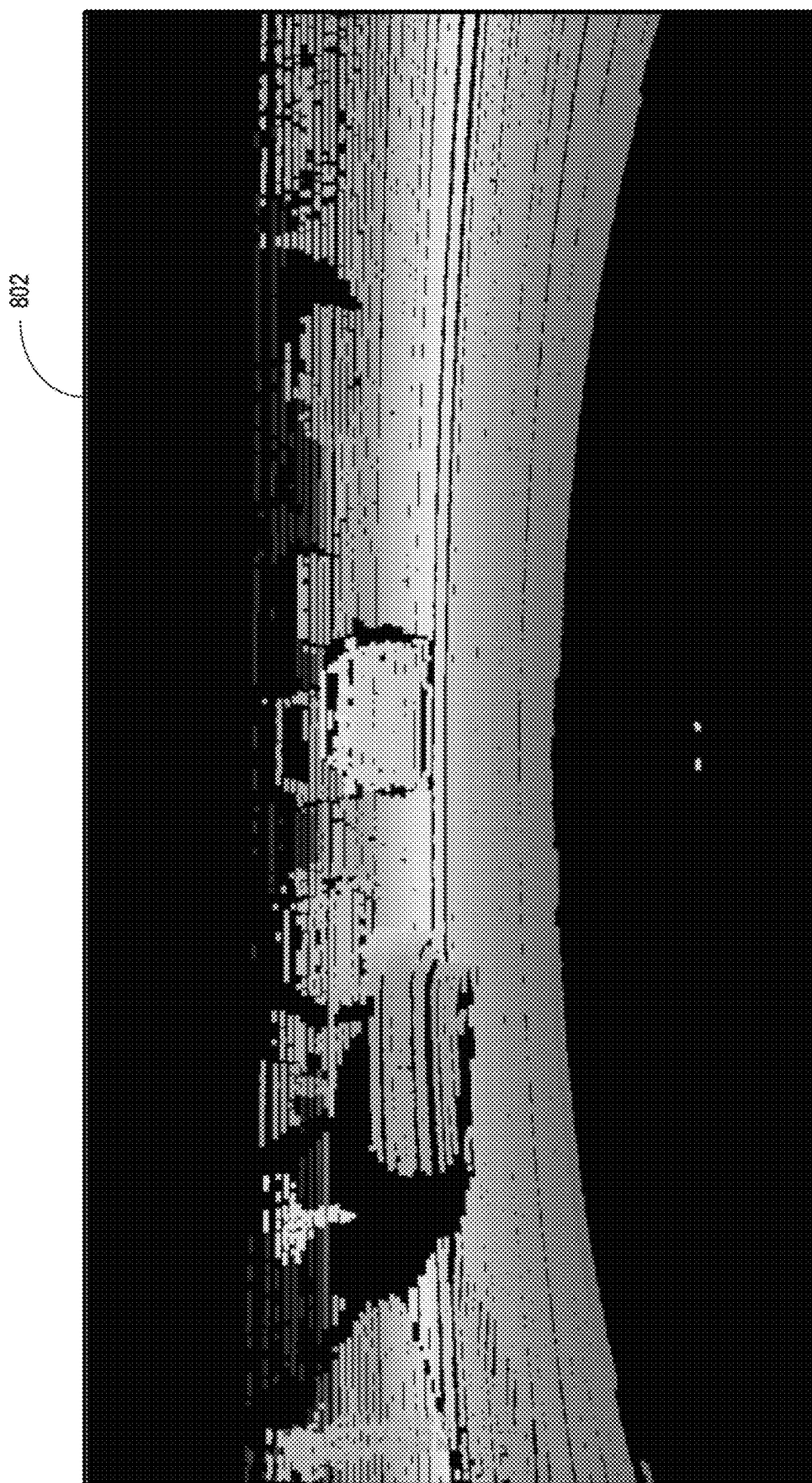


FIG. 8



FIG. 9

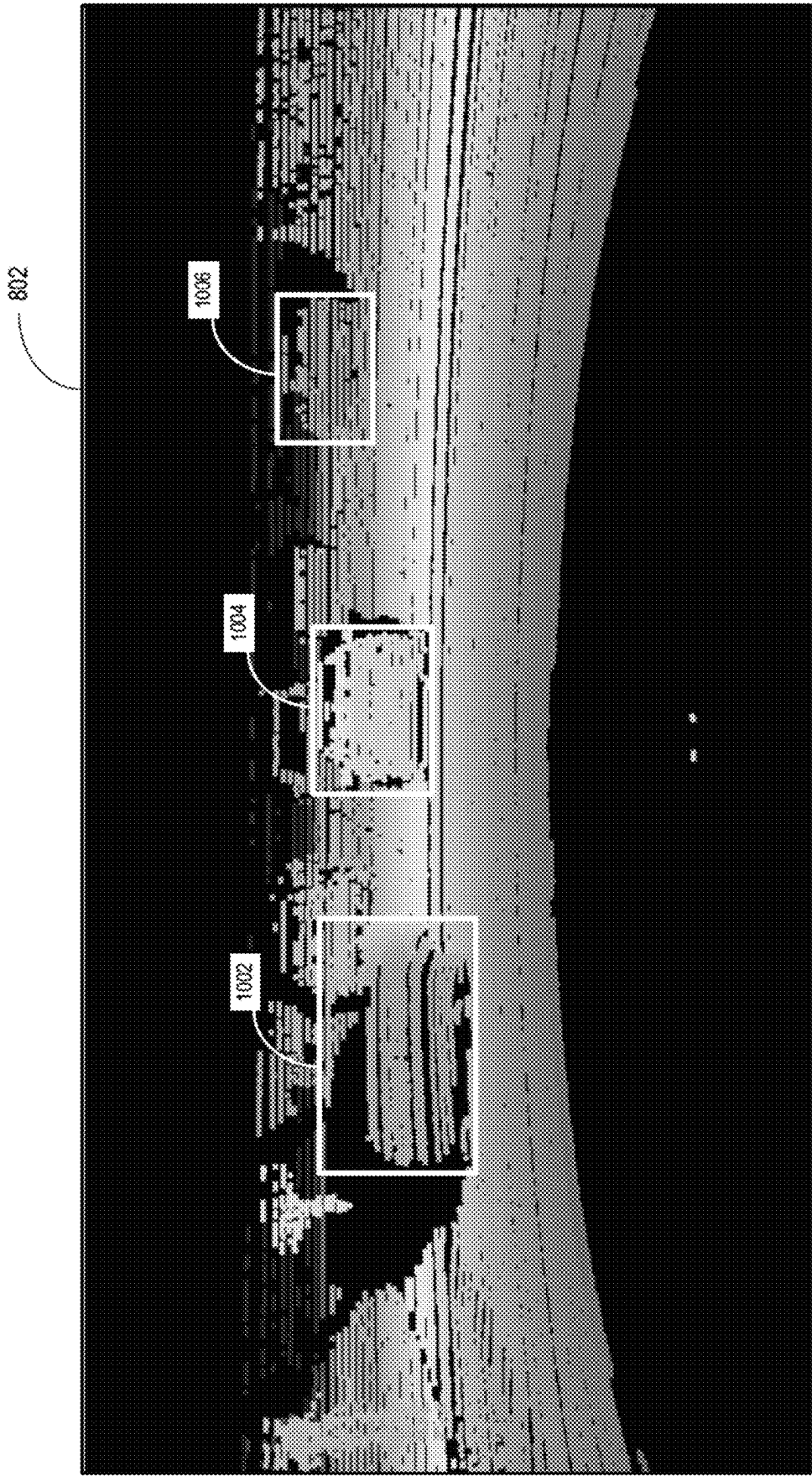


FIG. 10

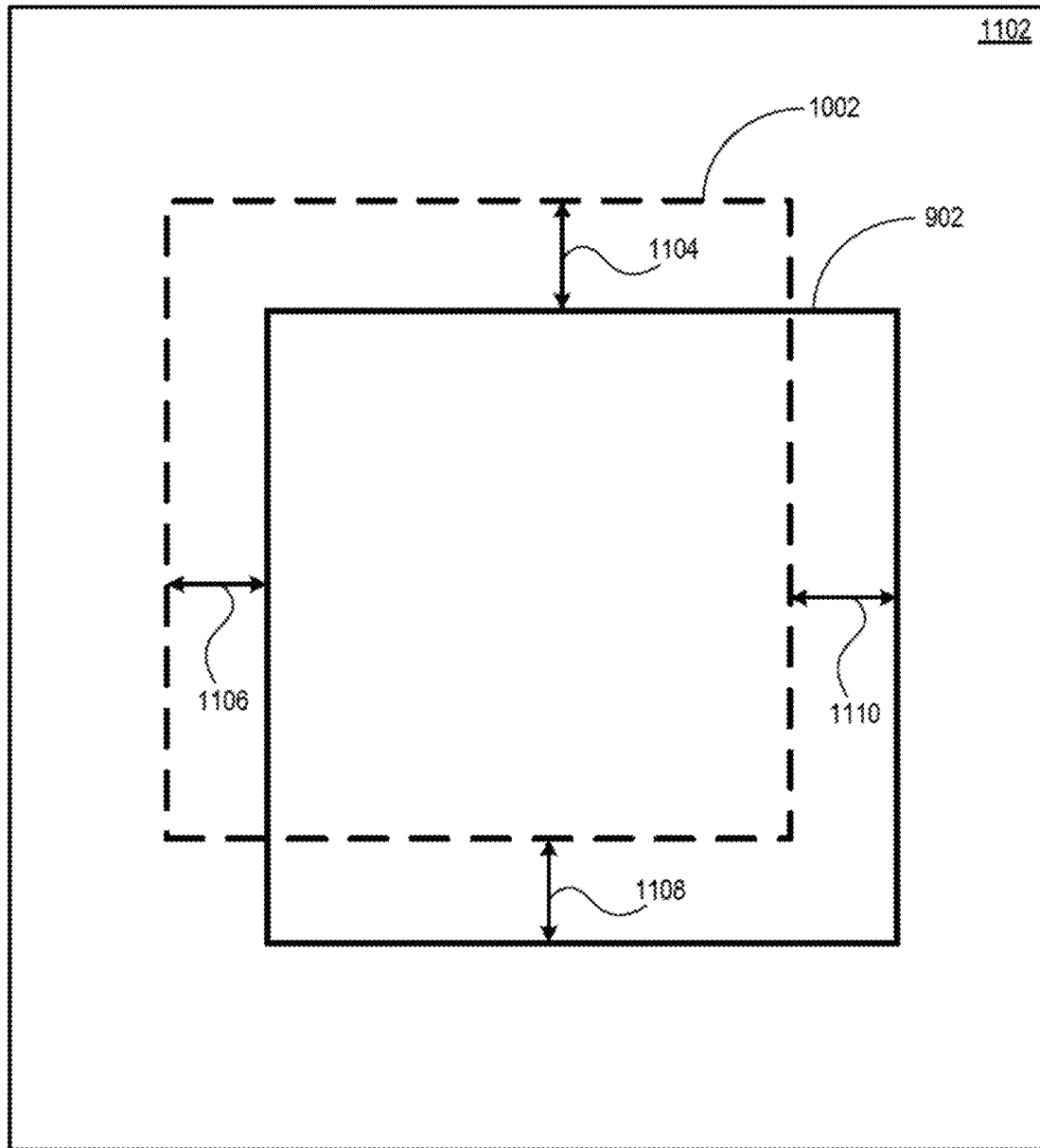


FIG. 11

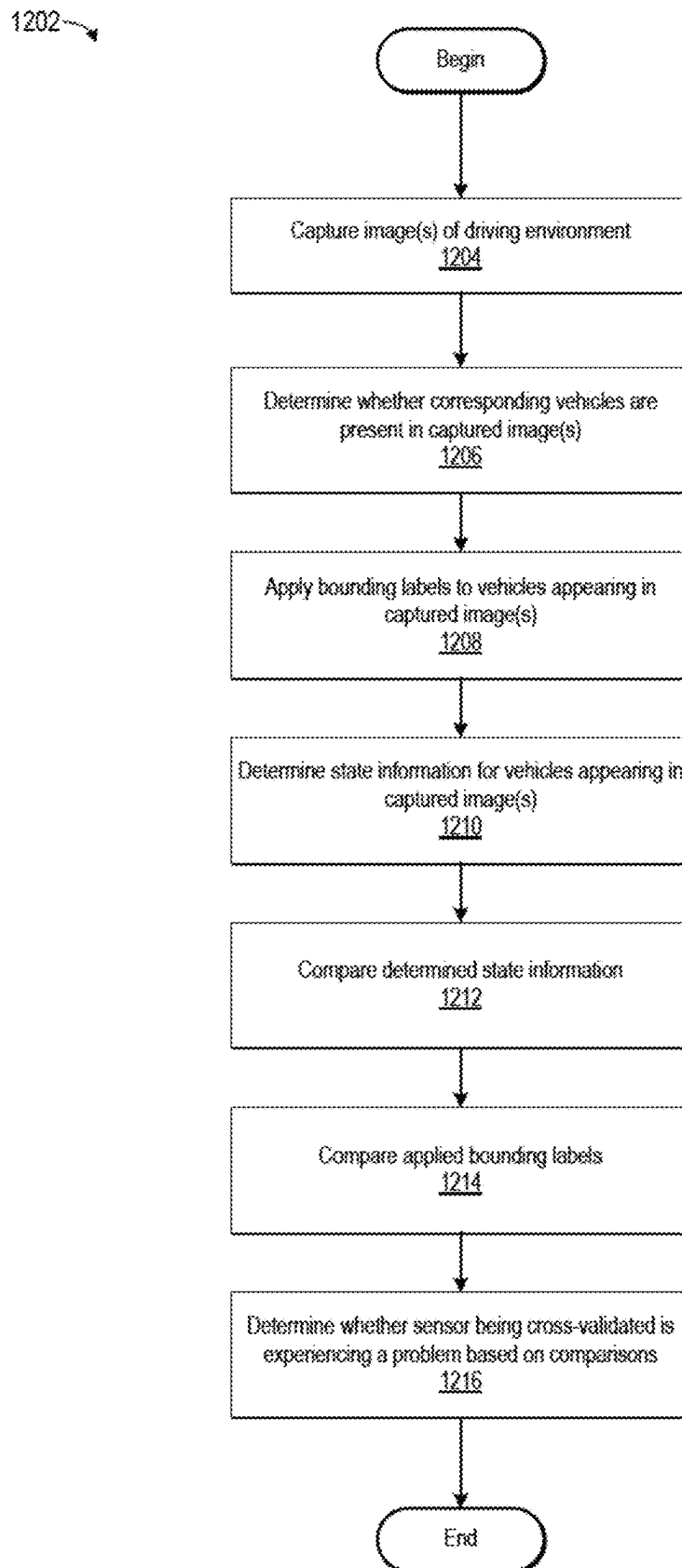


FIG. 12

CROSS-VALIDATING SENSORS OF AN AUTONOMOUS VEHICLE

CROSS-REFERENCE TO RELATED APPLICATIONS

The present application is a continuation of U.S. patent application Ser. No. 17/982,892, filed Nov. 8, 2022, which is a continuation of U.S. patent application Ser. No. 15/825,674, filed Nov. 29, 2017, now issued as U.S. Pat. No. 11,518,395, which is a continuation of U.S. patent application Ser. No. 15/383,198, filed Dec. 19, 2016, now issued as U.S. Pat. No. 9,868,446, which is a continuation of U.S. patent application Ser. No. 14/946,967, filed Nov. 20, 2015, now issued as U.S. Pat. No. 9,555,740, which is a continuation of U.S. patent application Ser. No. 13/628,986, filed Sep. 27, 2012, now issued as U.S. Pat. No. 9,221,396, the disclosures of which are incorporated herein by reference.

BACKGROUND

An autonomous vehicle may use various computing systems to aid in the transport of passengers from one location to another. In addition, the autonomous vehicle may require an initial input or continuous input from an operator, such as a pilot, driver, or passenger. Other autonomous systems, for example autopilot systems, may be used only when the system has been engaged, which permits the operator to switch from a manual mode (where the operator exercises a high degree of control over the movement of the autonomous vehicle) to an autonomous mode (where the autonomous vehicle essentially drives itself) to modes that lie somewhere in between.

The autonomous vehicle may be equipped with various types of sensors in order to detect objects in its environment. For example, the autonomous vehicles may include such sensors as lasers, sonar, radar, cameras, and other sensors that scan and record data from the autonomous vehicle's environment. Sensor data from one or more of these sensors may be used to detect objects and their respective characteristics (position, shape, heading, speed, etc.). This detection and identification is a critical function for the safe operation of the autonomous vehicle.

To navigate an environment confidently and precisely, the autonomous vehicle may rely on a prior stored electronic representation of the environment (e.g., a roadway, a highway, etc.). The electronic representation of the environment may be considered a "map" that identifies such features as lane markings, lane edges, k-rail concrete barriers, lane dividers, road medians, traffic safety cones, and other such features. The autonomous vehicle may store the map for both complex and simple environments.

However, the sensors on the autonomous vehicle may experience one or more problems, such as failure, inaccurate readings, or other such problems. When the sensor has a problem, the sensor may be unusable for determining whether there are objects, such as other vehicles, proximate to the autonomous vehicle. Having a sensor failure decreases the ability of the autonomous vehicle to navigate its environment confidently and precisely.

BRIEF SUMMARY

An apparatus and method are disclosed. In one embodiment, the apparatus includes a first sensor configured to detect a vehicle in a driving environment, a second sensor configured to detect the vehicle in the driving environment,

and a memory that stores detailed map information for the driving environment, first sensor parameters for determining whether the vehicle is present in a first image of the driving environment, and second sensor parameters for determining whether the vehicle is present in a second image of the driving environment. The apparatus may also include a processor in communication with the memory, the first sensor and the second sensor, the processor being configured to receive first sensor information from the first sensor, the first sensor information being used to produce the first image and receive second sensor information from the second sensor, the second sensor information being used to produce the second image. The processor may also be configured to determine whether the vehicle is in both the first image and the second image, determine a deviation value by comparing the first image with the second image when it is determined that the vehicle is present in the first image and the second image, and compare the determined deviation value with a deviation threshold to determine whether the first or second sensor has a problem detecting the vehicle.

In another embodiment of the apparatus, the first sensor is a first type of imaging sensor and the second sensor is a second type of imaging sensor, the first type of imaging sensor being different than the second type of imaging sensor.

In a further embodiment of the apparatus, the processor is further configured to select the first sensor as a reference sensor for determining whether the second sensor has the problem detecting the vehicle.

In yet another embodiment of the apparatus, the processor selects the first sensor based on a predetermined time interval.

In yet a further embodiment of the apparatus, the processor selects the second sensor for cross-validation with the first sensor based on an amount of time that has elapsed since the second sensor was last cross-validated.

In another embodiment of the apparatus, the memory further stores a multi-level failure threshold comprising a plurality of levels, wherein each level is associated with an action for the processor to perform, and a level of the plurality of levels is associated with a condition of the deviation value exceeding the deviation threshold. The processor may also perform the action of the level associated with the condition of the deviation value exceeding the deviation threshold when the processor determines that the deviation value exceeds the deviation threshold.

In a further embodiment of the apparatus, the memory stores a failure count threshold that identifies an action to be performed when the failure count threshold is exceeded, the comparison of the determined deviation value with the deviation threshold causes the failure count threshold to be exceeded, and the action identified by the failure count threshold is performed.

In yet another embodiment of the apparatus, the first image comprises a plurality of images determined from the first sensor information, the first sensor information being received during a predetermined time interval, and the second image comprises a plurality of images determined from the second sensor information, the second sensor information being received during the predetermined time interval. The processor may also be configured to determine first state information about the vehicle from the first plurality of images, determine second state information about the vehicle from the second plurality of images, and compare the first state information with the second state information to determine the deviation value.

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In yet a further embodiment of the apparatus, the first state information comprises at least one of a first vehicle speed, a first vehicle position, or a first vehicle heading, the second state information comprises at least one of a second vehicle speed, a second vehicle position, or a second vehicle heading, and the processor compares at least one of the first vehicle speed, first vehicle position or first vehicle heading with at least one of the second vehicle speed, second vehicle position, or second vehicle heading.

In another embodiment of the apparatus, the first image comprises a first object label encompassing the vehicle, the first object label comprising a first parameter that defines a characteristic of the first object label, the second image comprises a second object label encompassing the vehicle, the second object label comprising a second parameter that defines a characteristic of the second object label, and the processor compares the first image with the second image to determine the deviation value by comparing the first object label with the second object label.

In yet another embodiment of the apparatus, the processor compares the first object label with the second object label by comparing the first parameter with the second parameter.

In one embodiment of the method, the method includes receiving, with a processor in communication with a first sensor, first sensor information, the first sensor information being used to produce a first image of a driving environment, receiving, with the processor in communication with the second sensor, second sensor information, the second sensor information being used to produce the second image of the driving environment, and determining, with the processor, whether a vehicle is present in both the first image and the second image. The method may also include determining, with the processor, a deviation value by comparing the first image with the second image when it is determined that the vehicle is present in the first image and the second image, and comparing, with the processor, the determined deviation value with a deviation threshold to determine whether the first or second sensor has a problem detecting the vehicle.

In another embodiment of the method, the first sensor is a first type of imaging sensor and the second sensor is a second type of imaging sensor, the first type of imaging sensor being different than the second type of imaging sensor.

In a further embodiment of the method, the method includes selecting, with the processor, the first sensor as a reference sensor for determining whether the second sensor has the problem detecting the vehicle.

In yet another embodiment of the method, the method includes selecting, with the processor, the first sensor based on a predetermined time interval.

In yet a further embodiment of the method, the method includes selecting, with the processor, the second sensor for cross-validation with the first sensor based on an amount of time that has elapsed since the second sensor was last cross-validated.

In another embodiment of the method, the method includes determining whether the deviation value exceeds a deviation threshold associated with a level of a multi-level failure threshold, wherein an action for the processor to perform is associated with the level, and performing the action of the level based on the determination that the deviation value exceeds the deviation threshold.

In a further embodiment of the method, the comparison of the determined deviation value with the deviation threshold causes a failure count threshold to be exceeded, the failure count threshold identifying an action to be performed when

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the failure count threshold is exceeded. The method may also include performing the action identified by the failure count threshold.

In yet another embodiment of the method, the first image comprises a plurality of images determined from the first sensor information, the first sensor information being received during a predetermined time interval, and the second image comprises a plurality of images determined from the second sensor information, the second sensor information being received during the predetermined time interval. The method may also include determining, with the processor, the first state information about the vehicle from the first plurality of images, determining, with the processor, second state information about the vehicle from the second plurality of images, and comparing, with the processor, the first state information with the second state information to determine the deviation value.

In yet a further embodiment of the method, the first state information comprises at least one of a first vehicle speed, a first vehicle position, or a first vehicle heading, and the second state information comprises at least one of a second vehicle speed, a second vehicle position, or a second vehicle heading. The method may also include comparing, with the processor, at least one of the first vehicle speed, first vehicle position or first vehicle heading with at least one of the second vehicle speed, second vehicle position, or second vehicle heading.

In another embodiment of the method, the first image comprises a first object label encompassing the vehicle, the first object label comprising a first parameter that defines a characteristic of the first object label, and the second image comprises a second object label encompassing the vehicle, the second object label comprising a second parameter that defines a characteristic of the second object label. The method may also include comparing, with the processor, the first image with the second image to determine the deviation value by comparing the first object label with the second object label.

In a further embodiment of the method, comparing the first object label with the second object label comprises comparing the first parameter with the second parameter.

BRIEF DESCRIPTION OF THE DRAWINGS

The accompanying drawings are not intended to be drawn to scale. In the drawings, each identical or nearly identical component that is illustrated in various figures is represented by a like numeral. For purposes of clarity, not every component may be labeled in every drawing. In the drawings:

FIG. 1 illustrates an example of an autonomous vehicle configured to cross-validate one or more sensors of the autonomous vehicle according to aspects of the disclosure.

FIG. 2 illustrates an example of an interior of the autonomous vehicle according to aspects of the disclosure.

FIG. 3 illustrates an example of the placement of the one or more sensors of the autonomous vehicle according to aspects of the disclosure.

FIGS. 4A-4D illustrate various views of the approximate sensor fields of the various sensors of the autonomous vehicle according to aspects of the disclosure.

FIG. 5 illustrates an example of detailed map information that may be stored by the autonomous vehicle in accordance with aspects of the disclosure.

FIG. 6 illustrates an example of the autonomous vehicle detecting one or more vehicles in the driving environment according to aspects of the disclosure.

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FIG. 7 illustrates an example of a raw camera image captured by a camera sensor of the autonomous vehicle according to aspects of the disclosure.

FIG. 8 illustrates an example of a laser point cloud image captured by the laser sensor of the autonomous vehicle according to aspects of the disclosure.

FIG. 9 illustrates examples object labels being applied to detected vehicles shown in the raw camera image of FIG. 7 according to aspects of the disclosure.

FIG. 10 illustrates object labels being applied to detected vehicles shown in the laser point cloud image of FIG. 8 according to aspects of the disclosure.

FIG. 11 illustrates an example of comparing object labels applied to objects detected by different types of sensors according to aspects of the disclosure.

FIG. 12 illustrates an example of logic flow for cross-validating sensors of the autonomous vehicle according to aspects of the disclosure.

DETAILED DESCRIPTION

This disclosure provides for systems and methods for cross-validating one or more sensors of an autonomous vehicle. In particular, this disclosure provides for an autonomous vehicle that evaluates whether a sensor is accurately detecting an object (i.e., the sensor being cross-validated) based on the detection of that object by another sensor (i.e., a reference sensor). Using the reference sensor to evaluate whether a second sensor is operating as expected is referred to as “cross-validation.” In this regard, the reference sensor may be a different type of sensor than the second sensor (i.e., the sensor being evaluated).

In evaluating the second sensor, the autonomous vehicle may, in a first embodiment, determine one or more object parameter values from the reference sensor having detected an object, determine one or more expected object parameter values for the second sensor based on the determined object parameter values from the reference sensor, and then compare the expected object parameter values with the object parameter values (if any) determined from the second sensor having detected the object.

In a second embodiment, the autonomous vehicle may compare a first object label applied to the detected object by the autonomous vehicle based on sensor data received from the reference sensor, and then compare the first object label with a second object label applied to the detected object by the autonomous vehicle based on sensor data received from the second sensor. In this regard, either embodiment, or a combination thereof, may facilitate the cross-validation of one or more sensors of the autonomous vehicle.

FIG. 1 illustrates an autonomous vehicle 102 configured to cross-validate one or more of its sensors. The autonomous vehicle 102 may be configured to operate autonomously, e.g., drive without the assistance of a human driver. Moreover, the autonomous vehicle 102 may be configured to detect various objects and determine whether the detected object is a vehicle.

While certain aspects of the disclosure are particularly useful in connection with specific types of vehicles, the autonomous vehicle 102 may be any type of vehicle including, but not limited to, cars, trucks, motorcycles, busses, boats, airplanes, helicopters, lawnmowers, recreational vehicles, amusement park vehicles, farm equipment, construction equipment, trams, golf carts, trains, and trolleys.

In one embodiment, the autonomous driving computer system 144 may include a processor 106 and a memory 108.

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The autonomous driving computer system 144 may also include other components typically present in a general purpose computer.

The memory 108 may store information accessible by the processor 106, such as instructions 110 and data 112 that may be executed or otherwise used by the processor 106. The memory 108 may be of any type of memory operative to store information accessible by the processor 106, including a computer-readable medium, or other medium that stores data that may be read with the aid of an electronic device. Examples of the memory 108 include, but are not limited, a hard-drive, a memory card, read-only memory (“ROM”), random-access memory (“RAM”), digital video disc (“DVD”), or other optical disks, as well as other write-capable and read-only memories. Systems and methods may include different combinations of the foregoing, whereby different portions of the instructions and data are stored on different types of media.

The instructions 110 may be any set of instructions that may be executed directly (such as machine code) or indirectly (such as scripts) by the processor 106. For example, the instructions 110 may be stored as computer code on the computer-readable medium. In that regard, the terms “instructions” and “programs” may be used interchangeably herein. The instructions 110 may be stored in object code format for direct processing by the processor 106, or in any other computer language including scripts or collections of independent source code modules that are interpreted on demand or compiled in advance. Functions, methods and routines of the instructions 110 are explained in more detail below.

The data 112 may be retrieved, stored, or modified by processor 106 in accordance with the instructions 110. For instance, although the disclosed embodiments not limited by any particular data structure, the data 112 may be stored in computer registers, in a relational database as a table having a plurality of different fields and records, XML documents, flat files, or in any computer-readable format. By further way of example only, image data may be stored as bitmaps comprised of grids of pixels that are stored in accordance with formats that are compressed or uncompressed, lossless (e.g., BMP) or lossy (e.g., JPEG), and bitmap or vector-based (e.g., SVG), as well as computer instructions for drawing graphics. The data 112 may comprise any information sufficient to identify the relevant information, such as numbers, descriptive text, proprietary codes, references to data stored in other areas of the same memory or different memories (including other network locations) or information that is used by a function to calculate the relevant data.

The processor 106 may be any conventional processor, including Reduced Instruction Set Computing (“RISC”) processors, Complex Instruction Set Computing (“CISC”) processors, or combinations of the foregoing. Alternatively, the processor may be a dedicated device such as an applicant-specific integrated circuit (“ASIC”).

Although FIG. 1 functionally illustrates the processor 106, the memory 108, and other elements of the autonomous driving computer system 144 as being within the same block, it will be understood by those of ordinary skill in the art that the processor 106 and the memory 108 may actually comprise multiple processors and memories that may or may not be stored within the same physical housing. For example, the memory 108 may be a hard drive or other storage media located in a housing different from that of the autonomous driving computer system 144.

Accordingly, references to a processor or computer will be understood to include references to a collection of

processors or computers or memories that may or may not operate in parallel. Rather than using a single processor to perform the acts described herein, some of the components, such as steering components and deceleration components, may each have their own processor that only performs calculations related to the component's specific function.

In various embodiments described herein, the processor **106** may be located remotely from the autonomous vehicle **102** and may communicate with the autonomous vehicle wirelessly. In other aspects, some of the processes described herein are executed on a processor disposed within the autonomous vehicle **102** and others by a remote processor, including taking the acts necessary to execute a single maneuver.

The autonomous driving computer system **144** may include all of the components normally used in connection with a computer, such as a central processing unit (CPU), a memory (e.g., RAM and internal hard drives) storing data **112** and instructions such as an Internet browser or other software application, an electronic display **122** (e.g., a monitor having a screen, a small liquid crystal display ("LCD") touch-screen or any other electrical device that is operable to display information), one or more user input devices (e.g., a mouse, keyboard, touch screen and/or microphone), as well as various sensors (e.g., a video camera) for gathering the explicit (e.g., a gesture) or implicit (e.g., "the person is asleep") information about the states and desires of a person.

The autonomous vehicle **102** may also include a geographic position component **136** in communication with the autonomous driving computer system **144** for determining the geographic location of the autonomous vehicle **102**. For example, the geographic position component **136** may include a Global Positioning System ("GPS") receiver to determine the autonomous vehicle's **102** latitude, longitude and/or altitude position. Other location systems such as laser-based localization systems, inertial-aided GPS, or camera-based localization may also be used to identify the location of the vehicle. The location of the autonomous vehicle **102** may include an absolute geographical location, such as latitude, longitude, and altitude as well as relative location information, such as location relative to other vehicles immediately around it, which can often be determined with less noise than absolute geographical location.

The geographic position component **136** may also include other devices in communication with the autonomous driving computer system **144**, such as an accelerometer, gyroscope or another direction/speed detection device **138** to determine the direction and speed of the vehicle or changes thereto. By way of example only, the geographic position component **136** may determine its pitch, yaw or roll (or changes thereto) relative to the direction of gravity or a plane perpendicular thereto. The geographic position component **136** may also track increases or decreases in speed and the direction of such changes. The location and orientation data as set forth herein may be provided automatically to the user, the autonomous driving computer **144**, the vehicle central processor **126**, other computers and combinations of the foregoing.

The autonomous driving computer system **144** may control the direction and speed of the autonomous vehicle **102** by controlling various components. By way of example, if the autonomous vehicle **102** is operating in a completely autonomous mode, the autonomous driving computer system **144** may cause the autonomous vehicle **102** to accelerate via the acceleration system **130** (e.g., by increasing fuel or other energy provided to the engine), decelerate via the

braking system **128** (e.g., by decreasing the fuel supplied to the engine or by applying brakes) and change direction (e.g., by turning the front two wheels). The autonomous driving computer system **144** may also control one or more systems, such as the signaling system **130**, when controlling the acceleration system **130** and/or the braking system **128**.

The autonomous driving computer system **144** may also control one or more status indicators **118**, which may convey the status of the autonomous vehicle **102** and its components to a passenger. For example, the autonomous vehicle **102** may be equipped with an electronic display **122** for displaying information relating to the overall status of the vehicle, particular sensors, or information about or from the autonomous driving computer system **144**. The electronic display **122** may display computer-generated images of the vehicle's surroundings including, for example, the status of the autonomous driving computer system **144**, the autonomous vehicle **102** itself, roadways, intersections, as well as other objects and information.

The autonomous driving computer system **144** may use visual or audible cues to indicate whether it is obtaining valid data from one or more sensors, whether a passenger or the autonomous driving computer system **144** is partially or completely controlling the direction or speed of the autonomous vehicle **102** or both, such as whether there are any errors, etc. In addition, the autonomous driving computer system **144** may also have external indicators which indicate whether, at the moment, a human or an automated system is in control of the vehicle, that are readable by humans, other computers, or both.

The autonomous driving computer system **144** may also communicate with other components of the autonomous vehicle **102**. For example, the autonomous driving computer system **144** may communicate with a vehicle central processor **126**. The autonomous driving computer system **144** may also send and receive information from the various systems of the autonomous vehicle **102**. Communicating with the various systems may include communicating with the braking system **128**, the acceleration system **130**, the signaling system **132**, and the vehicle navigation system **134**. Communications with these systems may facilitate the control of the movement, speed, etc. of the autonomous vehicle **102**. In addition, when engaged, autonomous driving computer system **144** may control some or all of these functions of the autonomous vehicle **102** and thus be fully or partially autonomous. Although various systems and the autonomous driving computer system **144** are shown within the autonomous vehicle **102**, these systems and components may be external to the autonomous vehicle **102** or physically separated by large distances.

The autonomous vehicle **102** may also include one or more sensors **146** for detecting objects external to it, such as other vehicles, obstacles in the roadway, traffic signals, signs, trees, etc. The sensors **146** may include lasers, sonar, radar, cameras or any other detection devices. For example, where the autonomous vehicle **102** is a small passenger car, the small passenger car may include a laser mounted on the roof or other convenient location. In one aspect, the laser may measure the distance between the autonomous vehicle **102** and the object surfaces facing the autonomous vehicle **102** by spinning on its axis and changing its pitch. The autonomous vehicle **102** may also include various radar detection units, such as those used for adaptive cruise control systems. The radar detection units may be located on the front and back of the car as well as on either side of the front bumper. In another example, a variety of cameras may be mounted on the autonomous vehicle **102** at known

distances from one another. In this manner, the parallax from the different images may be used to compute the distance to various objects captured by the one or more cameras. These sensors may assist the vehicle in responding to its environment to maximize safety for passengers as well as objects or people in the environment.

FIG. 3 illustrates one example of the autonomous vehicle 102 and the placement of its one or more sensors. The autonomous vehicle 102 may include lasers 302 and 304, for example, mounted on the front and top of the autonomous vehicle 102, respectively. The laser 302 may have a range of approximately 150 meters, a thirty degree vertical field of view, and approximately a thirty degree horizontal field of view. The laser 304 may have a range of approximately 50-80 meters, a thirty degree vertical field of view, and a 360 degree horizontal field of view. The lasers 302-304 may provide the autonomous vehicle 102 with range and intensity information that the processor 106 may use to identify the location and distance of various objects. In one aspect, the lasers 302-304 may measure the distance between the vehicle and object surfaces facing the vehicle by spinning on its axes and changing their pitch. Other lasers with different ranges and fields of view may also be used.

The autonomous vehicle 102 may also include various radar detection units, such as those used for adaptive cruise control systems. The radar detection units may be located on the front and back of the car as well as on either side of the front bumper. As shown in the example of FIG. 3, the autonomous vehicle 102 includes radar detection units 306-312 located on the side (only one side being shown), front and rear of the vehicle. Each of these radar detection units 306-312 may have a range of approximately 200 meters for an approximately 18 degree field of view as well as a range of approximately 60 meters for an approximately 56 degree field of view. Again, other radar detection units with different ranges and fields of view may also be used.

In another example, a variety of cameras may be mounted on the autonomous vehicle 102. The cameras may be mounted at predetermined distances so that the parallax from the images of two or more cameras may be used to compute the distance to various objects. As shown in FIG. 3, the autonomous vehicle 102 may include two cameras 314-316 mounted under a windshield 318 near the rear view mirror (not shown). The camera 314 may include a range of approximately 200 meters and an approximately 30 degree horizontal field of view, while the camera 316 may include a range of approximately 100 meters and an approximately 60 degree horizontal field of view. Other cameras with different ranges and fields of view may also be used.

Each sensor may be associated with a particular sensor field defined by the ranges and fields of view for which the sensor may be used to detect objects. FIG. 4A is a top-down view of the approximate sensor fields of the various sensors. FIG. 4B depicts the approximate sensor fields 402 and 404 for the lasers 302 and 304, respectively based on the fields of view for these sensors. In this example, the sensor field 402 includes an approximately 30 degree horizontal field of view for approximately 150 meters, and the sensor field 404 includes a 360-degree horizontal field of view for approximately 80 meters.

FIG. 4C depicts the approximate sensor fields 406-420 and for radar detection units 306-312, respectively, based on the fields of view for these sensors. For example, the radar detection unit 306 includes sensor fields 406 and 408. The sensor field 406 includes an approximately 18 degree horizontal field of view for approximately 200 meters, and the

sensor field 408 includes an approximately 56 degree horizontal field of view for approximately 80 meters.

Similarly, the radar detection units 308-312 include the sensor fields 410/414/418 and sensor fields 412/416/420. The sensor fields 410/414/418 include an approximately 18 degree horizontal field of view for approximately 200 meters, and the sensor fields 412/416/420 include an approximately 56 degree horizontal field of view for approximately 80 meters. The sensor fields 410 and 414 extend passed the edge of FIGS. 4A and 4C.

FIG. 4D depicts the approximate sensor fields 422-424 of cameras 314-316, respectively, based on the fields of view for these sensors. For example, the sensor field 422 of the camera 314 includes a field of view of approximately 30 degrees for approximately 200 meters, and sensor field 424 of the camera 316 includes a field of view of approximately 60 degrees for approximately 100 meters.

In general, an autonomous vehicle 102 may include sonar devices, stereo cameras, a localization camera, a laser, and a radar detection unit each with different fields of view. The sonar may have a horizontal field of view of approximately 60 degrees for a maximum distance of approximately 6 meters. The stereo cameras may have an overlapping region with a horizontal field of view of approximately 50 degrees, a vertical field of view of approximately 10 degrees, and a maximum distance of approximately 30 meters. The localization camera may have a horizontal field of view of approximately 75 degrees, a vertical field of view of approximately 90 degrees and a maximum distance of approximately 10 meters. The laser may have a horizontal field of view of approximately 360 degrees, a vertical field of view of approximately 30 degrees, and a maximum distance of 100 meters. The radar may have a horizontal field of view of 60 degrees for the near beam, 30 degrees for the far beam, and a maximum distance of 200 meters. Hence, the autonomous vehicle 102 may be configured with any arrangement of sensors having different fields of view, ranges, and sensor fields, and each of these sensors may capture one or more raw images for detecting the various objects near and around the autonomous vehicle 102.

In addition to the sensors 146 described above, the autonomous driving computer system 144 may also use input from sensors found in non-autonomous vehicles. As examples, these sensors may include tire pressure sensors, engine temperature sensors, brake heat sensors, break pad status, tire tread sensors, fuel sensors, oil level and quality sensors, air quality sensors (for detecting temperature, humidity, or particulates in the air), etc.

The data provided by the sensors 146 may be processed by the autonomous driving computer system 144 in real-time. In this context, the sensors 146 may continuously update their output to reflect the environment being sensed at or over a range of time, and continuously or as demanded. The sensors 146 may provide the updated output to the autonomous driving computer system 144 so that it can determine whether the autonomous vehicle's 102 then-current direction or speed should be modified in response to the sensed environment.

Referring back to FIG. 1, the autonomous vehicle 102 may also include an electronic representation of a driving environment for maneuvering in the driving environment, and for determining whether there are one or more objects proximate to the autonomous vehicle 102 in the driving environment. For example, autonomous driving computer system 144 may include detailed map information 114 that defines one or more driving environments. The detailed map information 114 may include various maps that identify the

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shape and elevation of roadways, lane lines, intersections, crosswalks, speed limits, traffic signals, buildings, signs, real time traffic information, or other such objects and information. The detailed map information **114** may further include explicit speed limit information associated with various roadway segments. The speed limit data may be entered manually or scanned from previously taken images of a speed limit sign using, for example, optical-character recognition. In addition, the detailed map information **114** may include three-dimensional terrain maps incorporating one or more of the objects (e.g., crosswalks, intersections, lane lines, etc.) listed above.

The detailed map information **114** may also include lane marker information identifying the location, elevation, and shape of lane markers. The lane markers may include features such as solid or broken double or single lane lines, solid or broken lane lines, reflectors, etc. A given lane may be associated with left and right lane lines or other lane markers that define the boundary of the lane. Thus, most lanes may be bounded by a left edge of one lane line and a right edge of another lane line.

The autonomous vehicle **102** may also include persistent data for detecting objects and cross-validating one or more of the sensors **146** based on the detected objects. For example, the data **112** may include one or more sensor parameters **148** used by the processor **106** to determine when a sensor has detected an object, and the type of object detected by the sensor. More particularly, the sensor parameters **148** may define an arrangement of pixels, laser points, intensity maps, etc., should be considered an object. The sensor parameters **148** may also define how an object is classified.

Each of the sensors **146** of the autonomous vehicle **102** may be associated with a corresponding set of sensor parameters **148**. Thus, the one or more camera(s) may be associated with camera parameters, the one or more laser(s) may be associated with laser parameters, and the one or more radar detection unit(s) may be associated with radar parameters. Examples of camera parameters may include the minimal brightness of a pedestrian, the minimum pixel size of a car object, the minimum width of a car object, and other such parameters. Examples of laser parameters may include the height of a pedestrian, the length of a car object, an obstacle detection threshold, and other such parameters. Examples of radar parameters may include minimum distance to an object, a delay threshold for detecting an object, the height of a pedestrian, and other such parameters.

In detecting vehicles in various driving environments, the data **112** may include vehicle data **116** that defines one or more parameters for classifying a vehicle. Classifications of vehicle may include such classifications as “passenger car,” “bicycle,” “motorcycle,” and other such classifications. The parameters defined by the vehicle data **116** may inform the autonomous driving computer system **144** as to the type of vehicle detected by a given sensor. For example, the vehicle data **116** may include parameters that define the type of vehicle when the vehicle is detected by one or more of the camera sensors, one or more of the laser sensors, and so forth.

Vehicles may be identified through a vehicle detection algorithm **124**, which the processor **106** may use to classify vehicles based on various characteristics, such as the size of the vehicle (bicycles are larger than a breadbox and smaller than a car), the speed of the vehicle (bicycles do not tend to go faster than 40 miles per hour or slower than 0.1 miles per hour), and other such characteristics. In addition, the vehicle may be classified based on specific attributes of the vehicle,

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such as information contained on a license plate, bumper sticker, or logos that appear on the vehicle.

The vehicle data **116** may also include state, positional, and/or trajectory information collected by the autonomous driving computer system **144** when a vehicle is detected. The autonomous driving computer system **144** may collect the state and/or positional information about a detected object such as a vehicle to assist in the determination of the detected vehicle’s trajectory. The vehicle trajectory for the detected vehicle may define the direction and speed that a vehicle has when in a given driving environment. The vehicle trajectory may also define the past positions, directions, and speed that the detected vehicle had while in the driving environment.

The autonomous vehicle **102** may generate state information about detected objects, such as other vehicles, regardless of whether the autonomous vehicle **102** is operating in an autonomous mode or a non-autonomous mode. Thus, whether the autonomous vehicle **102** is operating by itself or has a driver, the autonomous vehicle **102** may collect state and object information, for example, in order to determine the aforementioned vehicle trajectories.

The instructions **110** may also include a cross-validation algorithm for cross-validating one or more of the sensors **146**. As previously discussed, cross-validating a second sensor (i.e., the sensor being validated) may be performed by detecting an object with a first sensor, and then determining whether the second sensor also detects the object and/or the accuracy of the second sensor in detecting the object.

The autonomous driving computer system **144** may take various actions defined by the cross-validation algorithm **140** in response to the cross-validation operations previously described. For example, the autonomous driving computer system **144** may display a warning to a passenger of the autonomous vehicle **102** when the autonomous driving computer system **144** determines that the sensor being cross-validated is experiencing a problem (e.g., not accurately reporting a detected object). The warning may inform the passenger which sensor is not operating within operational parameters. As another example, the autonomous driving computer system **144** may request that a passenger take control of the autonomous vehicle **102**. Moreover, the autonomous driving computer system **144** may perform a combination of actions, such as displaying the warning and requesting that a passenger take control of the autonomous vehicle **102**.

FIG. 2 illustrates an example of an interior of the autonomous vehicle **102** according to aspects of the disclosure. The autonomous vehicle **102** may include all of the features of a non-autonomous vehicle, for example: a steering apparatus, such as steering wheel **210**; a navigation display apparatus, such as navigation display **215**; and a gear selector apparatus, such as gear shifter **220**. The vehicle **102** may also have various user input devices, such as gear shifter **220**, touch screen **217**, or button inputs **219**, for activating or deactivating one or more autonomous driving modes and for enabling a driver or passenger **290** to provide information, such as a navigation destination, to the autonomous driving computer **106**.

The autonomous vehicle **102** may also include one or more additional displays. For example, the autonomous vehicle **102** may include a display **225** for displaying information regarding the status of the autonomous vehicle **102** or its computer. In another example, the autonomous vehicle **102** may include a status indicating apparatus such as status bar **230**, to indicate the current status of vehicle **102**. In the example of FIG. 2, the status bar **230** displays

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“D” and “2 mph” indicating that the autonomous vehicle 102 is presently in drive mode and is moving at 2 miles per hour. In that regard, the autonomous vehicle 102 may display text on an electronic display, illuminate portions of the autonomous vehicle 102, such as the steering wheel 210, or provide various other types of indications.

FIG. 5 illustrates an example of a portion of a detailed map 502 that may represent the driving environment of the autonomous vehicle 102. The detailed map 502 may be retrieved or referenced by the autonomous vehicle 102 based on a detected position of the autonomous vehicle 102. The detailed map 502 may be stored as part of the detailed map information 114.

The detailed map 502 may further represent a section of a road, such as highway, parkway, etc., and may include lane information such as information about solid lane lines 504-508 and broken lane lines 510-512. These lane lines may define lanes 514-520. Each of the lanes 514-520 may be associated with a respective centerline rail 522-528 which may indicate the direction in which a vehicle should generally travel in the respective lane. For example, a vehicle may follow centerline rail 522 when driving in lane 514. In this example, the lane 514 may be bounded by a left lane line 504 and the right lane line 506. Similarly, the lane 516 may be bounded by the left lane line 506 and the right lane line 508, the lane 518 may be bounded by the left lane line 508 and the right lane line 510, and the lane 520 may be bounded by the left lane line 510 and the right lane line 512.

Each of the lanes 514-520 may be bounded by corresponding lane edges. Thus, lane 514 may be bounded by edges 530,532, lane 516 may be bounded by edges 534,536, lane 518 may be bounded by edges 538,540 and lane 520 may be bounded by edges 542,544.

In the example shown in FIG. 5, the detailed map information 114 may be depicted as an image-based map. However, the detailed map information 114 need not be entirely or completely image-based (e.g., raster-based). For example, the detailed map information 114 may include one or more roadgraphs or graph networks of information such as roads, lanes, intersections, and the connections between these features. Each feature may be stored as graph data and may be associated with information such as a geographic location and whether or not it is linked to other related features, for example, a stop sign may be linked to a road and an intersection, etc. In some examples, the associated data may include grid-based indices of a roadgraph to allow for efficient lookup of certain roadgraph features.

The detailed map information 114 may be loaded into the memory 108 of the autonomous vehicle 102 at a predetermined time. In one embodiment, the detailed map information 114 may be loaded into the memory 108 of the autonomous vehicle 102 on a daily basis. Alternatively, or in addition, the detailed map information 114 may be loaded into the memory 108 at other predetermined times, such as on a monthly or weekly basis.

As previously discussed, the autonomous driving computer system 144 may include instructions 110 having various algorithms for detecting objects in a driving environment and for cross-validating one or more of the sensors 126 based on the detected objects.

The vehicle detection algorithm 124 may facilitate the detection of vehicles by the autonomous driving computer system 144. FIG. 6 illustrates an example 602 of the autonomous vehicle 102 in a driving environment. The detailed map 502 may correspond to the driving environment of FIG. 6. In the example 602, the autonomous vehicle 102 may detect one or more vehicles, such as vehicles

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604-608, using the vehicle detection algorithm 124. The vehicles 604-608 may be within range of one or more of the sensor fields projected by the autonomous vehicle 102, such as the laser sensor field 404 and the camera sensor field 424. The vehicles 604-608 may be within range of other sensor fields, but these other sensor fields have been omitted for simplicity and clarity.

In the example of FIG. 6, the vehicles 604-608 and the autonomous vehicle 102 may be in various lanes of the driving environment. In particular, the vehicle 604 may be in the lane 514, the vehicle 606 may be in the lane 516, and the vehicle 608 may be in the lane 520. The autonomous vehicle 102 may be in the lane 516.

The autonomous driving computer system 144 may receive data from sensors regarding various objects in the autonomous vehicle's surroundings. For example, the autonomous driving computer system 144 may receive data regarding the vehicles 604-608 from the laser 304 (corresponding to the laser sensor field 404) and from the one or more of the cameras 314/316 (corresponding to the camera sensor field 404). FIG. 7 illustrates a camera image 702 captured using one or more of the cameras 314/316. FIG. 8 illustrates an example of a laser point cloud image 802 generated using data from the laser 304. The view shown in the laser point cloud image 802 may approximate the view shown in the camera image 702.

In one embodiment of the cross-validation algorithm 140, the autonomous driving computer system 144 may determine which of the sensors to cross-validate, i.e., which of the sensors to evaluate. In one example, the autonomous driving computer system 144 may determine that the laser 304 is to be cross-validated. As noted above the autonomous driving computer system 144 may determine to cross-validate the laser 304 based on such factors as time (e.g., when the laser 304 was last cross-validated), distance (e.g., how far the autonomous vehicle 104 travelled since last cross-validating the laser 304), manual request (e.g., a passenger requests cross-validation of the laser 304), or any other such factors or combination of factors.

The autonomous driving computer system 144 may then select a first sensor by which to cross-validate the sensor determined for validation above. For example, the autonomous driving computer system 144 may select one or more of the cameras 314/316 to perform the cross-validation of the laser 304. Selection of the cameras 314/316 may be based on one or more selection criterion, such as when the cameras 314/316 were used in a cross-validation or cross-validated themselves, whether the cameras 314/316 are operating within expected parameters, time between cross-validations, and other such criterion.

In cross-validating a sensor, the autonomous driving computer system 144 may be configured to cross-validate the sensor based on one or more parameters. For example, the autonomous vehicle computer system 144 may be configured to cross-validate one or more of the sensors 126 at predetermined time intervals, such as every thirty seconds, every minute, every ten minutes, and so forth. As another example, the autonomous vehicle computer system 144 may be configured to cross-validate one or more of the sensors 126 based on distance travelled by the autonomous vehicle 102, such as every ten meters, every one hundred meters, and so forth. Moreover, different sensors 126 may be cross-validated at different intervals [you have time and distance in the example], e.g., a camera sensor may be cross-validated every ten minutes and a laser sensor may be cross-validated every ten meters.

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In cross-validating the sensors of the autonomous vehicle **102**, the autonomous driving computer system **144** may use the objects detected in the images produced from the various sensor **14**. For example, the autonomous driving computer system **144** may perform vehicle detection on the captured camera image **702** according to the vehicle detection algorithm **124**. In this regard, the autonomous driving computer system **144** may refer to one or more of the sensor parameters **148**, such as the camera parameters previously discussed, to determine whether there are objects present in the captured camera image **702**, and which of the objects are vehicles.

When the autonomous driving computer system **144** detects an object, the autonomous driving computer system **144** may apply an object label to the detected object based on one or more of the sensor parameters **148** previously discussed. The object label may be defined by a bounding box encompassing the object. In alternative embodiments, the object label may be defined by a bounding oval or other bounding shape. FIGS. **9-10** illustrate examples of object labels being applied to various objects (i.e., vehicles) detected in the camera image **702** and the laser point cloud image **802**.

An object label may have one or more object label parameters that define the shape of the object label. The object label parameters may be defined by one or more of the instructions **110**, such as the vehicle detection algorithm **124**. Moreover, the object label parameters may vary depending on the sensor type of the sensor that captured the image. For example, where the shape of the object label is a bounding box, and the sensor that captured the image is a one or more of the cameras, the object label parameters may include a height parameter that defines the height of the bounding box (e.g., in pixels), a width parameter that defines the width of the bounding box (e.g., in pixels), a first pixel coordinate that defines the latitudinal placement of the bounding box (e.g., an X-coordinate), and a second pixel coordinate that defines the longitudinal placement of the bounding box (e.g., a Y-coordinate).

Where the sensor that captured the image is one or more of the lasers, the object label parameters may also include a third pixel coordinate that defines the physical height of the object or a particular laser point depicted in the captured image (e.g., a Z-coordinate). This third pixel coordinate should not be confused with the height parameter of the object label, because this third pixel coordinate may indicate the elevation of the detected object or of a given laser point (e.g., 3 meters above sea level, 2 meters above sea level, etc.) This third pixel coordinate may further indicate the height of the detected object or laser point relative to the autonomous vehicle **102**.

Where the sensor that provides data about the object is one or more of the radar detection units, the object label parameters may include a height parameter that defines the height of the height of the bounding box (e.g., in pixels), a width parameter that defines the width of the bounding box (e.g., in pixels), a first pixel coordinate that defines the latitudinal placement of the bounding box (e.g., an X-coordinate), and a second pixel coordinate that defines the longitudinal placement of the bounding box (e.g., a Y-coordinate).

FIG. **9** illustrates an example of object labels **902-906** having been applied to vehicles detected in the camera image **702** based on the vehicle detection algorithm **124**. As previously discussed, the object labels **902-906** may take the form of a rectangular bounding box that encompasses a detected object. In FIG. **9**, the object label **902** may be

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applied to the vehicle **604**, the object label **904** may be applied to the vehicle **606**, and the object label **906** may be applied to the vehicle **608**.

The autonomous driving computer system **144** may store various parameters about the object labels **902-906**. These parameters may include two-dimensional coordinates for each of the vertices of the object labels **902-906**, the width of each of the object labels **902-906**, the height of each of the object labels **902-906**, and other such parameters.

Having detected the vehicles **604-608**, the autonomous driving computer system **144** may also determine state information for each of the vehicles **604-608**. State information may include characteristics about the detected vehicles **604-608**. Examples of state information include, but are not limited to, the detected vehicle's speed, the path traveled by the vehicle, the driving environment in which the vehicle is traveling, any directional or orientation changes by the vehicle, or other such state information. The state information may also be associated with one or more segments of the detailed map information **114** to further refine the state of the detected vehicle. For example, where the detected vehicle is detected as being on a highway (as defined by the detailed map information **114**), the collected state information may identify that the detected vehicle was traveling in a highway, and may further identify the direction of the detected vehicle, various positional information or changes about the detected vehicle (e.g., the original starting lane of the detected vehicle, the ending lane of the detected vehicle), and other such state information.

In determining state information for one or more of the vehicles **604-608**, the autonomous driving computer system **144** may determine the state information based on one or more state information parameters. Determining state information for a vehicle may be based on a number of state information parameters associated with the sensor data including predetermined time intervals, distance traveled by the autonomous vehicle **102**, number of images produced by the sensor data, clarity and/or accuracy of the images, and other such factors.

For example, with regard to time, the autonomous driving computer system **144** may determine the state information for a vehicle based on one or more camera images, laser point cloud images, radar intensity images, and so forth, produced over a period of two seconds, five seconds, one minute, or other measurement of time.

With regard to distance, the autonomous driving computer system **144** may determine the state information for a vehicle based on one or more images produced over a distance of five meters, ten meters, one kilometer, or other measurement of distance.

Similarly, with regard to the number of images, the autonomous driving computer system **144** may determine the state information for a vehicle based on two images, three images, fifteen images, or any other number of images. Combinations of the state information parameters being used in the determination of the state information for a vehicle are also possible.

State information for one or more of the detected vehicles **604-608** may be used to cross-validate the laser **304**. To determine state information for one or more of the vehicles **604-608**, the autonomous driving computer system **144** may track and/or monitor the vehicles **604-608** as they travel in the driving environment. In this regard, the state information determined about each of the detected vehicles **604-608** may include the vehicle's position, the vehicle's speed, the vehicle's direction of travel, and other such information. In one embodiment, the autonomous driving computer system

144 may determine this state information from additional camera images (not shown) captured by the one or more cameras 314/316 (e.g., through one or more extrapolation techniques).

The autonomous driving computer system 144 may also determine whether any of the vehicles 604-608 are expected in the field of view of the sensor being cross-validated (i.e., the laser 304). The autonomous driving computer system 144 may perform this determination by determining the position of the vehicles 604-608 relative to the detailed map information 114, and then extrapolating whether the field of view of the laser 304 (i.e., within the sensor field 404) would encompass the vehicles' positions. The vehicles that the autonomous driving computer system 144 determines would be within the field of view of the laser 304 may be used to cross-validate the laser 304.

For example, the autonomous driving computer system 144 may determine that each of the vehicles 604-608 are within the field of view of the laser 304. More generally, the autonomous driving computer system 144 may exclude objects having been detected by the sensor being used in the cross-validation (i.e., the reference sensor), but not within the field of view of the sensor being cross-validated (i.e., the sensor being evaluated). In this regard, the autonomous driving computer system 144 may ensure that only objects appearing in the field of view for both the reference sensor and the sensor being evaluated are used in the cross-validation.

In one embodiment, cross-validation of a sensor may be performed by leveraging the state information determined from the sensor readings of the reference sensor. In general, the autonomous driving computer system 144 may determine vehicle state information based on sensor readings from the reference sensor, and then compare the vehicle state information determined from sensor readings of the reference sensor with vehicle state information determined from the sensor readings of the sensor being evaluated. With reference to the cameras 314/316 (i.e., the reference sensor (s)) and the laser 304 (i.e., the sensor to be evaluated), the autonomous driving computer system 144 may determine vehicle state information based on the sensor readings from one or more of the cameras 314/316, determine vehicle state information based on the sensor readings from the laser 304, and then compare the state information determined from the one or more cameras 314/316 with the state information determined from the laser 304. Based on this comparison, the autonomous driving computer system 144 may determine whether the laser sensor 304 is operating within operational parameters. Similar comparisons may be performed between other sensors used by the computer system 144 to detect and respond to objects external to the autonomous vehicle 102.

In another embodiment, the autonomous driving computer system 144 may apply a first object label to a detected object for a first sensor, apply a second object label to the detected object for a second sensor, and then compare the applied first and second object labels. This may include one or more comparisons of various parameters associated with each object label, such as height and/or width comparisons, positional placement comparisons, vertex coordinate comparisons, object classification comparisons, and other such comparisons. Based on these one or more comparisons, the autonomous driving computer system 144 may determine whether the second sensor detected the object, and whether an accurate object label was applied to the detected object.

FIG. 10 illustrates an example of object labels 1002-1006 having been applied to the vehicles 604-608 detected by the

laser 304. The autonomous driving computer system 144 has applied an object label 1002 to the vehicle 604, an object label 1004 to the vehicle 606, and an object label to the vehicle 1006.

The autonomous driving computer system 144 may determine which objects should receive object labels based on the laser sensor parameters previously discussed with reference to FIG. 1. In addition, the autonomous driving computer system 144 may identify disturbances and/or differences in the captured laser point cloud image 802 by comparing the captured laser point cloud image 802 with an expected laser point cloud image based on the detailed map information 114. Areas of the captured laser point cloud image 802 that are different from the expected laser point cloud image based on the detailed map information 114 may indicate an object present in the captured laser point cloud image 802. Those areas may then be analyzed by the autonomous driving computer system 144 to determine whether an object is present in that area. The autonomous driving computer system 144 may then classify the object based on the stored vehicle data 116.

As with determining state information for the vehicles 604-606 based on one or more captured camera images, the autonomous driving computer system may determine state information for the vehicles 604-606 based on one or more laser point cloud images by comparing consecutive captured laser point cloud images. The autonomous driving computer system 144 may then store the state information determined from the laser point cloud images as part of the vehicle data 116.

Determining state information from the sensor data of two different sensors may be based on the same or different state information parameters. For example, the state information parameters used in determining the state information based on the laser point cloud images may be the same, or different from, the state information parameters used in determining state information for the vehicles based on the captured camera images. In this regard, determining state information based on the laser point cloud images may be based on laser point cloud images captured over a distance of two meters, whereas determining state information based on the camera images may be based on camera images captured over a period of two minutes. The values of the state information parameters may also vary (e.g., a time period state information parameter for the camera may be thirty seconds and a time period state information parameter for the laser sensor may be every minute).

The autonomous driving computer system 144 may then compare the state information determined from the determined sensor data with the state information determined from the sensor data from the first sensor selected to cross-validate the determined sensor. Moreover, the cross-validation algorithm 140 may contain instructions such that the autonomous driving computer system 144 compares state information for only objects that appear both in the field of view of the one or more cameras 314/316 and in the field of view of the laser 304.

Comparing state information for one or more vehicles may include comparing any number of vehicle characteristics. For example, comparing state information may include a comparing vehicle's position, a vehicle's speed, a vehicle's heading (i.e., direction of travel), the size of a vehicle, the type of vehicle, or any other combination of characteristics. Furthermore, the autonomous driving computer system 144 may compare the state information relative to the detailed map information 114 (e.g., the detailed map 502) and/or relative to the autonomous vehicle 102. For example,

the autonomous driving computer system **144** may compare the position of the vehicle **604** determined from one or more camera images with the position of the vehicle **604** determined from one or more laser point cloud images, relative to the autonomous vehicle **102** or relative to the detailed map **502**. Similarly, the autonomous driving computer system **144** may compare the speed of the vehicle **604** determined from one or more camera images with the speed of the vehicle **604** determined from one or more laser point cloud images, relative to the autonomous vehicle **102** or relative to the detailed map **502**. Comparing the state information may also be in terms absolute values.

In one embodiment, the results of comparing the state information from the various sensors should indicate that the state information obtained from the sensors is substantially, or nearly, identical. Otherwise, deviations (i.e., differences) in the state information may indicate that a sensor (i.e., the laser **304**) is experiencing a problem. To determine whether there are differences, and the extent of those differences, the autonomous driving computer system **144** may employ one or more state information thresholds. For example, there may be a positional threshold (e.g., one meter), a speed threshold (e.g., 0.5 km/hr), a size threshold (e.g., height and/or width), a classification threshold (e.g., one size classification above or below), and other such state information thresholds.

In addition to the state information thresholds, the autonomous driving computer system **144** may establish a multi-level failure threshold based on the compared state information. The multi-level failure threshold may include escalating levels of failure that indicate the extent to which a sensor being cross-validated (i.e., laser **304**) is experiencing a problem. For example, the multi-level failure threshold may have a first level of one threshold difference, a second level of two threshold differences, a third level of three threshold differences, and so forth. Thus, where the compared state information indicates that the state information position threshold has been exceeded, the autonomous driving computer system **144** may record that the first level of the multi-level failure threshold was reached. Similarly, where the compared state information indicates that the state information position threshold and the state information speed threshold have been exceeded, the autonomous driving computer system **144** may record that the second level of the multi-level failure threshold was reached.

Based upon the level of failure reached in the multi-level failure threshold, the autonomous driving computer system **144** may take a variety of actions. For example, at the first level of the multi-level failure threshold, the autonomous driving computer system **144** may record in an electronic log that the first level was reached. At the second level of the multi-level failure threshold, the autonomous driving computer system **144** may display an error message. At the third level, the autonomous driving computer system **144** may request that a passenger take control of the autonomous vehicle **102**. In this manner, the multi-level failure threshold serves as a safety mechanism to alert a passenger when there is a problem with a sensor being cross-validated.

The autonomous driving computer system **144** may also maintain a total failure count threshold. The total failure count threshold may be in addition to the multi-level failure threshold. The total failure count threshold may be a threshold based on the number of times a state information threshold has been exceeded, either during the cross-validation of a sensor, the lifetime of a sensor, or at any other time. There may be a global total failure count threshold (i.e., a failure count threshold for all the sensors **146** of the

autonomous vehicle **102**), total failure count thresholds for each sensor of the autonomous vehicle **102**, or combinations thereof. Furthermore, each of the state information characteristics (e.g., position, speed, heading, etc.) may each have their own failure count threshold. The failure count threshold may be as low as five (or lower) or as high as ten (or higher).

Similar to the multi-level failure threshold, the autonomous driving computer system **144** may take a variety of actions based on the value of the total failure count threshold (global or individual sensor), based on the values of the individual failure count thresholds for the state information, or a combination thereof. For example, each time the individual sensor total failure count is increased, the autonomous driving computer system **144** may record this fact in an electronic log. However, when the individual sensor total failure count is equal to or greater than a first predetermined value (e.g., fifty), the autonomous driving computer system **144** may display an error message regarding the failure (e.g., the fact that a sensor failed to cross-validate, that a specific state information characteristic was different, etc.). Moreover, when the individual sensor total failure count is equal to or greater than a second predetermined value (e.g., two hundred), the autonomous driving computer system **144** may request that a passenger take control of the autonomous vehicle **102**.

Each of the failure thresholds (e.g., the multi-level failure threshold and/or the failure count threshold(s)), may be used to quickly identify different problems in the sensors of the autonomous vehicle **102**. For example, the multi-level failure threshold may be used to identify a single, catastrophic failure in a sensor (e.g., the failure count may be low, but all of the state information thresholds may have been exceeded). As another example, a failure count threshold may be used to identify an intermittent problem, such as where a failure count threshold is increased at time t_1 (e.g., one minute from time t_0), but then increased again at time t_2 (e.g., two minutes from time t_2). In this regard, the various failure thresholds for the compared state information serve as safety mechanisms to readily identify potential failures or problems in sensors being cross-validated.

As mentioned above, comparing determined state information may encompass one embodiment of the cross-validation algorithm **140**. In another embodiment of the cross-validation algorithm **140**, the autonomous driving computer system **144** may be configured to compare the object labels applied to the objects detected in the captured images (e.g., the camera image **702** and the laser point cloud image **802**).

FIG. **11** illustrates an example **1102** of comparing a first object label (e.g., object label **902**) and a second object label (e.g., object label **1002**). The autonomous driving computer system **144** may compare various characteristics of the object labels **902,1002**. For example, the autonomous driving computer system **144** may compare the height and/or width of the object labels **902,1002**, the placement of the object labels **902,1002** in the camera image **702** and the laser point cloud image **802**, the individual locations of the vertices of the object labels **902,1002** relative to their respective images **702,802**, and other such characteristics.

FIG. **11** illustrates comparing the height and width of the object labels **902,1002**. When corrected for perspective (e.g., the laser point cloud image **802** may be slightly skewed compared to the camera image **702**), a comparison of the height and width of the object labels **902,1002** may serve as a proxy for determining whether the laser **304** is operating within expected parameters.

In comparing the object labels **902,1002**, the autonomous driving computer system **144** may determine a number of object label deviations. The number of object label deviations may be dependent on the number of sides of the object label. For example, where the object label is a polygon, such as a rectangular bounding box, the object label deviations may include four deviations: a first height deviation **1104**, a second height deviation **1108**, a first width deviation **1106**, and a second width deviation **1110**. The number of deviations may be more where there are more sides to the object labels, such as in the case of pentagons, hexagons, or other multi-sided polygons, or the number of deviations may be less, such as in the case of triangles, circles, ovals, or other similar shapes.

The object label deviations may be measured in physical units (e.g., centimeters, meters, etc.) or in electronic units (e.g., pixels). In this regard, the deviations may be assigned values based on the physical characteristics of the object encompassed by the object label (e.g., vehicle **604**) or the deviations may be assigned values based on the characteristics of the captured image in which the object appears.

In addition to determining object label deviations based on individual comparisons (e.g., the comparison of object label **902** with the object label **1002**), the autonomous driving computer system **144** may also determine deviations based on an aggregation of comparisons, such as ten, twenty, or more comparisons. In this regard, the comparisons may include a first comparison of a first object label applied to the vehicle **604** in a first image captured by the one or more cameras **314/316** with a first object label applied to the vehicle **604** in a first image captured by the laser **304**, a second comparison of a second object label applied to the vehicle **604** in a second image captured by the one or more cameras **314/316** with a second object label applied to the vehicle **604** in a second image captured by the laser **304**, and so forth.

The object label deviations determined from the aggregation of comparisons may include one or more maximum deviations, one or more average deviations, and one or more average signed deviations. For example, there may be four maximum deviations, four average deviations, and four average signed deviations (i.e., a total of twelve deviations), where each deviation corresponds to a side of a rectangular bounding box, such as the object labels **902, 1002**. There may be more or less maximum, average, and signed average deviations depending on the number of sides of the object labels being compared.

The values of the maximum deviation(s), average deviation(s), and/or average signed deviation(s) may vary. In particular, the value of the maximum deviation may equal the value of the maximum difference for a compared set of deviations (e.g., thirty pixels, twenty pixels, fifty meters, etc.). The value of the average deviation may equal the average value of the differences for a compared set of deviations (e.g., two pixels, one meter, etc.). The value of the average signed deviation may equal the average value of the differences for a compared set of deviations and may also indicate whether a side of a object label is shifted to the left/down (where the average signed deviation is negative) or shifted to the right/up (where the average signed deviation is positive). Examples of values for the average signed deviation may include -10 pixels, +10 pixels, -2 meters, +2 meters, and other such values. The maximum, average, and average signed deviations may be based on all sides being compared, for a single side being compared, or combinations thereof.

In one embodiment, the autonomous driving computer system **144** may establish one or more deviation thresholds by which to compare the aforementioned deviations, including the object label deviations **1104-1110**. In this regard, the autonomous driving computer system **144** may establish one or more deviation thresholds for a singular object label (e.g., a first height object label deviation threshold) and/or one or more deviation thresholds for the determined maximum deviation, the determined average deviation, and the determined average signed deviation.

Comparisons of the deviations with their corresponding deviation thresholds may indicate whether the sensor being cross-validated is experiencing a problem and the extent of that problem. For example, a comparison of the maximum deviation with the maximum deviation threshold may indicate the magnitude of the problem being experienced by the sensor. Similarly, a comparison of the average signed deviation with the average signed deviation threshold may, not only indicate the magnitude of the problem being experienced by the sensor, but also the extent to which the sensor may be improperly calibrated (e.g., are the sensor readings too far to the left, too far to the right, too high up, too far down, etc.).

The values of the one or more deviation thresholds may be different, or the same as, each other. For example, a first width deviation threshold may have the same value as a second width deviation threshold. However, the values may also be different. Similarly, a first height deviation threshold may have the same or different value as a second height deviation threshold.

Combinations of the foregoing embodiments are also possible. For example, in comparing state information determined from the one or more cameras **314/316** with the state information determined from the laser **304**, the autonomous driving computer system **144** may establish a maximum deviation and maximum deviation threshold, an average deviation and an average deviation threshold, and an average signed deviation and an average signed deviation threshold. Similarly, in comparing object labels (e.g., object labels **902,1002**), the autonomous driving computer system **144** may establish a multi-level failure threshold for monitoring when one or more of the height and/or width deviation thresholds have been exceeded, and the autonomous driving computer system **144** may establish a failure count threshold for monitoring the number of times one or more of the deviation thresholds for the object labels **902,1002** have been exceeded.

FIG. **12** illustrates an example of logic flow **1202** for cross-validating a sensor of the autonomous vehicle **102**. Initially, one or more sensors **146** of the autonomous vehicle may capture one or more images of the driving environment (Block **1204**). As discussed with reference to FIG. **7** and FIG. **8**, these images may include camera images and/or laser point cloud images.

The autonomous driving computer system **144** may then select which sensor to use a reference sensor (e.g., one or more of the cameras **314/316**) and which sensor to cross-validate (e.g., the laser **304**). The autonomous driving computer system **144** may then determine whether, and which, objects appearing in one or more images captured by the reference sensor are in one or more images captured by the sensor to be cross-validated (Block **1206**). The autonomous driving computer system **144** may then apply object labels to the objects appearing in both sets of images (Block **1208**). Alternatively, the object labels may be applied to objects in the first and second set of images, and then the autonomous

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driving computer system **144** may determine which objects the first and second set of images have in common.

The autonomous driving computer system **144** may then determine state information from the one or more common objects appearing in the one or more images captured by the reference sensor and determine state information for the same objects appearing in the one or more images captured by the sensor being cross-validated (Block **1210**). As previously discussed, the state information may include position (relative to the autonomous vehicle **102** and/or driving environment), changes in position, speed (also relative to the autonomous vehicle **102** and/or driving environment), changes in speed, heading, and other such state information.

The autonomous driving computer system **144** may then compare the state information for the common objects to determine one or more deviation values (Block **1212**). As previously discussed, the deviation values may include deviation values specific to individual state characteristics (e.g., a positional deviation value, a speed deviation value, etc.) but may also include deviation values determined in the aggregate (e.g., a maximum deviation value, an average deviation value, and an average signed deviation value).

Alternatively, or in addition, the autonomous driving computer system **144** may compare the object labels applied to the common objects (Block **1214**). Comparing the object labels may include comparing the object labels from images captured at approximately the same time and/or comparing the object labels from one or more images captured during a predetermined time interval. As with the comparison of the state information, the autonomous driving computer system **144** may determine various deviation values based on the comparison of the object labels.

The autonomous driving computer system **144** may then determine whether the sensor being cross-validated is having a problem (Block **1216**). As discussed previously, the autonomous driving computer system **144** may perform this determination by comparing one or more of the determined deviation values with one or more deviation thresholds. The autonomous driving computer system **144** may also establish one or more failure thresholds (e.g., the multi-level failure threshold, the failure count threshold, etc.) that may be exceeded when one or more of the deviation values exceed their corresponding deviation thresholds. The autonomous driving computer system **144** may be configured to perform a variety of actions (e.g., record in an electronic log, display a warning message, request a passenger to take control of the autonomous vehicle, etc.) based on whether one or more of the failure thresholds are exceeded.

In this manner, the autonomous driving computer system **144** determines whether a sensor is experiencing a problem based on sensor readings obtained from another sensor (i.e., the reference sensor). The sensors may even be different types of sensors. Moreover, the autonomous driving computer system **144** may be configured to determine whether the sensor being cross-validated has experienced a catastrophic failure (e.g., the sensor readings from the sensor being cross-validated are substantially different than the sensor readings from the reference sensor) or is experiencing an intermittent problem. As the cross-validation methods and system disclosed herein may be used to determine when a sensor is experiencing a problem, the safety and confidence of the autonomous vehicle in navigating a driving environment is enhanced.

Although aspects of this disclosure have been described with reference to particular embodiments, it is to be understood that these embodiments are merely illustrative of the

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principles and applications of the present disclosure. It is therefore to be understood that numerous modifications may be made to the illustrative embodiments and that other arrangements may be devised without departing from the spirit and scope of this disclosure as defined by the appended claims. Furthermore, while certain operations and functions are shown in a specific order, they may be performed in a different order unless it is expressly stated otherwise.

The invention claimed is:

1. A method comprising:

receiving, by one or more processors, first sensor data captured by a first sensor configured to detect objects in an environment of the first sensor;

receiving, by the one or more processors, second sensor data captured by a second sensor configured to detect objects in the environment, the second sensor being of a different type than the first sensor;

identifying, by the one or more processors, an object from the first sensor data;

validating, by the one or more processors, accuracy of the first sensor by comparing a first classification of the object based on the first sensor data to a second classification of the object based on the second sensor data;

based on validating the accuracy of the first sensor, determining, by the one or more processors, whether a failure threshold has been met; and
when the failure threshold has been met, generating, by the one or more processors, an error message.

2. The method of claim 1, wherein the first sensor is a camera and the second sensor is a RADAR sensor.

3. The method of claim 1, wherein the first sensor is a RADAR sensor and the second sensor is a camera.

4. The method of claim 1, wherein the first sensor is a camera and the second sensor is a LIDAR sensor.

5. The method of claim 1, wherein the first sensor is a LIDAR sensor and the second sensor is a camera.

6. The method of claim 1, wherein the first classification corresponds to a motorcycle.

7. The method of claim 1, wherein the first classification corresponds to a bicyclist.

8. The method of claim 1, further comprising:

based on validating the accuracy of the first sensor, increasing a count value corresponding to a number of failures for the first sensor,

wherein determining whether the failure threshold has been met includes comparing the count value to a predetermined failure value.

9. The method of claim 1, further comprising, prior to the comparing, determining that the object appears in a field of view of both the first sensor and the second sensor.

10. A system comprising:

a first sensor configured to detect objects in an environment of the first sensor;

a second sensor configured to detect objects in the environment, the second sensor being of a different type than the first sensor; and

one or more processors configured to:

receive first sensor data captured by the first sensor;

receiving second sensor data captured by the second sensor;

identify an object from the first sensor data;

validate accuracy of the first sensor by comparing a first classification of the object based on the first sensor data to a second classification of the object based on the second sensor data;

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based on validating the accuracy of the first sensor,
determine whether a failure threshold has been met;
and

when the failure threshold has been met, generate an
error message.

11. The system of claim 10, wherein the first sensor is a
camera and the second sensor is a RADAR sensor.

12. The system of claim 10, wherein the first sensor is a
RADAR sensor and the second sensor is a camera.

13. The system of claim 10, wherein the first sensor is a 10
camera and the second sensor is a LIDAR sensor.

14. The system of claim 10, wherein the first sensor is a
LIDAR sensor and the second sensor is a camera.

15. The system of claim 10, wherein the first classification
corresponds to a motorcycle.

16. The system of claim 10, wherein the first classification 15
corresponds to a bicyclist.

17. The system of claim 10, wherein the one or more
processors are further configured to:

based on validating the accuracy of the first sensor, 20
increase a count value corresponding to a number of
failures for the first sensor; and

determine whether the failure threshold has been met by
being further configured to compare the count value to
a predetermined failure value.

18. The system of claim 10, wherein the one or more
processors are further configured to, prior to the comparing,

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determine that the object appears in a field of view of both
the first sensor and the second sensor.

19. A non-transitory computer-readable medium on which
instructions are stored, the instructions, when executed by
one or more processors, cause the one or more processors to
perform a method, the method comprising:

receiving first sensor data captured by a first sensor
configured to detect objects in an environment of the
first sensor;

receiving second sensor data captured by a second sensor
configured to detect objects in the environment, the
second sensor being of a different type than the first
sensor;

identifying an object from the first sensor data;

validating accuracy of the first sensor by comparing a first
classification of the object based on the first sensor data
to a second classification of the object based on the
second sensor data;

based on validating the accuracy of the first sensor,
determining whether a failure threshold has been met;
and

when the failure threshold has been met, generating an
error message.

20. The medium of claim 19, wherein the error message 25
includes a request for a human operator.

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